EOSS

PRODUCT REQUIREMENTS DOCUMENT

A manufacturing-first Business OS for solo founders and small teams

Easy data entry. Effortless data collection.
Intelligent automation. Streamlined daily operations.

> **PRODUCT NORTH STAR**
> EOSS should make running a business easier, not give a founder another system to manage. It captures information where work happens, connects it across the business and helps people act on what matters.

VERSION 1.3 | BASE PRD V1.0 + BINDING CLARIFICATIONS | 4 NOVEMBER 2026

Status: Consolidated working PRD with binding v1.1 product-owner clarifications

Initial vertical: Indian manufacturing / industrial fabrication

Scope baseline: 12 workspaces, 27 screen identifiers, 94 planned entities, approximately 920 fields, 37 reusable components and 12 demo roles (corrected from 10)

Document control and how to use this PRD

Field	Value
Product	EOSS — Enterprise Operational Solution Systems (product name used in the current prototype).
Document	Consolidated Product Requirements Document (PRD).
Version / date	v1.3 / 5 November 2026 (base v1.0 plus binding clarifications)
Status	Working product baseline for implementation; v1.3 clarifications supersede conflicting v1.0 statements.
Initial target	Indian MSME manufacturing / fabrication businesses, designed for solo founders and small teams.
Prototype baseline	Full-coverage interactive prototype across 12 workspaces and 27 screen identifiers; 94 planned entities, approximately 920 fields, 37 reusable components and 12 demo roles (corrected from 10).
Companion artifacts	Product Definition v0.1; Product Architecture & UX Blueprint; Definition Review; Implementation Clarification Addendum; Scope Matrix v1.1; field dictionary workbook.

Source hierarchy and reconciliation rules

- Current product-owner decisions and explicit decisions in this conversation govern where older documents conflict.
- The original EOSS Scope Matrix v1.1 baseline is retained as the historical source. This consolidated PRD now includes binding v1.3 clarifications: 12 workspaces, 27 screen identifiers, 94 planned entities, approximately 920 fields, 37 reusable components and 12 demo roles (corrected from 10). The exact field-level schema remains subject to Field Dictionary v0.2 reconciliation.
- The EOSS Pilot Product Architecture & UX Blueprint and Product Definition v0.1 are the primary manufacturing workflow and pilot references. Accepted corrections from the 51-gap Definition Review are reflected where reconciled in the later scope matrix.
- The Implementation Clarification Addendum governs dependency sequencing and the clarified permission principles; its original four-week demonstration milestone does not narrow the current full-coverage prototype objective.
- The Enterprise OS BRD v2.0 and Enterprise OS UX Blueprint v0.3 are broader cross-industry platform references. Their capabilities are recorded as strategic or future scope where they exceed the current manufacturing-first MVP.
- The Scope Matrix v1.1 refers to Field Dictionary v0.2, but the assembled source set available for this PRD contains Field Dictionary v0.1 only. Therefore, exact field-level sign-off must be reconciled against the actual v0.2 workbook before a production schema is frozen.

> **IMPORTANT DISTINCTION**
>
> This PRD separates (1) the full-coverage interactive prototype, (2) the first manufacturing pilot release and (3) the broader cross-industry Business OS vision. These are related, but they are not the same release.

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1. Executive summary and product definition

EOSS is a connected operational system for founders and small teams. The initial implementation focus is manufacturing, but the product philosophy is broader: capture work with minimal effort, connect the resulting data and automate routine operational follow-through.

1.1 Product definition

EOSS is an AI-native Business Operating System that helps a business owner and a small team manage day-to-day operations with less manual chasing, fewer disconnected records and clearer control over customers, suppliers, materials, jobs, production, quality, machines, documents and delivery.

For the initial manufacturing vertical, EOSS provides a job-connected operating layer from customer enquiry and order intake through procurement, goods receipt, inventory, production, quality release, dispatch and delivery confirmation. Inventory, Suppliers, Assets and Documents remain independently navigable first-class workspaces; they are not hidden inside a Job screen.

1.2 Product mantra

> **THE EOSS MANTRA**
>
> Easy data entry. Effortless data collection. Intelligent automation. Streamlined daily operations.

- Easy data entry: minimise typing, use smart defaults, prefill from known records, provide quick capture and progressive forms, and collect information at the point of work.
- Effortless data collection: capture information once and reuse it across workspaces. A GRN updates inventory; a stage update updates job progress; an inspection updates quality status; a dispatch updates delivery and finished-goods movement.
- Intelligent automation: detect exceptions, link affected records, draft follow-ups and recommend next actions. Use deterministic rules for calculations and control decisions; use AI for extraction, classification, summarisation and recommendations where it adds value.
- Streamlined operations: give the owner an immediate view of what needs attention, what is progressing, what is blocked, what can be automated and what requires a decision.

1.3 Product promise

EOSS reduces the effort needed to keep a business operationally informed and coordinated. It is not intended to force every company into a heavyweight ERP process or require the founder to become a full-time data-entry administrator.

1.4 Product positioning

Position EOSS as	Do not position EOSS as
A practical operating layer connecting work, records, evidence, people, suppliers, materials and decisions.	Another accounting package, a generic chatbot, or a dashboard-only product.
A manufacturing-first Business OS that can expand to additional business models over time.	A claim that incumbent ERPs cannot perform manufacturing, stock or procurement functions.
A product focused on low-effort adoption, connected context and actionable intelligence.	A replacement for every existing system on day one.

2. Vision, problem and product principles

2.1 Vision

Give solo founders and small teams the leverage of a coordinated operations function without requiring them to build a large administrative team. EOSS should progressively assemble the operational picture from the activities people already perform.

2.2 Problems to solve

Problem hypothesis	EOSS response
Operational information is scattered across email, WhatsApp, spreadsheets, paper records and business software.	Capture or import information into canonical business records, preserve source evidence and link related records.
Owners and supervisors spend time chasing status from employees, suppliers and customers.	Provide shared status, commitments, ownership, reminders and exception views grounded in records.
Material availability is difficult to reconcile with job requirements, stores and supplier commitments.	Show on-hand, reserved, quality-held, in-transit and available quantities with job and lot traceability.
Production delays are discovered late or lack a clear reason.	Track stages, quantities, blockers, machine dependencies and planned versus actual dates.
Certificates, drawings, inspection evidence and machine documents are difficult to find.	Provide a central Document Vault with versioning, expiry, access control and entity links.
Routine decisions and follow-ups consume founder attention.	Prepare recommendations and action drafts, route approvals and automate permitted reminders.

These are pilot problem hypotheses drawn from the 3 November factory discussion and discovery materials. Frequency, severity and willingness to pay must be validated with pilot users; they are not universal claims about all Indian manufacturers.

2.3 Product principles

Principle	Requirement
Business-first, not AI-first	Every core task remains possible through normal navigation, structured forms, lists, search and filters. AI is a layer of intelligence, not the only interface.
Job-connected, not job-only	The Job connects commercial orders, requirements, inventory, procurement, production, quality, assets and dispatch. Other domains remain first-class and independently accessible.
Operational adoption over feature count	Reduce data-entry burden, support assisted setup and imports, capture updates close to where work happens and avoid forcing unnecessary process steps.
Capture once, reuse everywhere	Use canonical IDs and linked records. Do not ask users to re-enter supplier, customer, material, job or document information already known.
Deterministic controls, assistive AI	Stock balances, permissions, approvals, status transitions and quantities use explicit rules. AI handles unstructured extraction, classification, explanation and recommendations.
AI proposes; humans confirm	No consequential business action is executed solely because an AI system proposed it. Require permission checks and explicit confirmation.
Make complexity progressive	Keep the common task simple. Reveal advanced fields and workflows when the business context requires them; do not hide important capabilities.
Configurable common patterns	Stage templates, material attributes, UOMs, inspection checklists, sites, store locations and role scopes should be configurable without customer-specific code forks.
Security is foundational	Tenant/site isolation, backend-enforced RBAC, document access control, encryption, audit and privacy are required before real customer data is loaded.

3. Target users, personas and jobs-to-be-done

3.1 Target-user levels

Level	Profile	Experience requirement
Level 1 — Solo owner	One owner, usually one site, with no dedicated functional team.	Owner-first Pulse, quick capture, simple workflows, low setup effort and clear next actions.
Level 2 — Owner with team	Owner plus a small team, approximately 1–10 employees as a planning persona.	Role-aware workspaces, delegated tasks, approvals, employee attribution and mobile task capture.
Level 3 — Multiple teams or sites	Owner coordinating multiple locations, teams or operational units.	Site-scoped work, multi-site stock and records, role/record scope, cross-site visibility when granted.

3.2 Primary personas

Persona	Primary responsibility	Main job-to-be-done
Founder / Owner / GM	Owns customer commitments, cash/stock risk and operational decisions.	Understand business health quickly; see the highest-value issues; approve exceptional actions; avoid chasing multiple people.
Sales Executive	Captures enquiries, prepares quotations and records customer POs.	Capture RFQs quickly; reuse customer/order context; preserve quote versions and drawing revisions; answer delivery questions.
Buyer / Procurement	Sources material, negotiates with suppliers and tracks commitments.	Turn shortages into purchase requests; compare offers; record supplier promises; identify overdue deliveries and affected jobs.
Storekeeper	Receives, stores, allocates and issues material.	Capture receipts and lots on phone/tablet; maintain stock accuracy; trace certificates; avoid duplicate entry.
Production Supervisor	Coordinates jobs, people, work centres and progress.	Know what can run, what is blocked and why; record stage quantities quickly; attribute work to employees.
Operator	Performs a specific stage or operation.	Complete an assigned stage with minimal typing; capture quantities, note and evidence; no unnecessary menus or approvals.
QC Inspector	Performs incoming, in-process and final checks.	Execute a checklist, capture measurements and evidence, raise NCRs and support controlled quality release.
Maintenance Technician	Handles breakdowns, preventive maintenance and return to service.	Report breakdown from mobile; link affected jobs; record work and spares; document downtime.
Dispatch Personnel	Prepares shipment, dispatch documents and delivery evidence.	Assemble job/PO lines, lots and documents; confirm dispatch; capture POD and delivered quantities.
Auditor	Has time-scoped, read-only access to permitted records and evidence.	Trace who did what, when and against which record; view permitted documents with logged access.

3.3 Cross-cutting usage realities

- One person may hold multiple responsibilities. Role experience must not assume one employee per function.
- Some shop-floor employees may not have an EOSS user account. Employee identity and application-user identity must be separate but linkable.
- Users may share a phone or tablet; the actual performer of a task must not be inferred automatically from the logged-in recorder.
- Mobile and low-bandwidth use are important for GRN, stage update, quality checklist, breakdown report, stock lookup and dispatch/POD capture.
- English is the initial UI language. The data model and UI architecture should avoid hard-coded assumptions that block future Indian-language labels and input.

4. Goals, outcomes and success measures

Success measures are to establish baselines during assisted setup and the pilot. No unvalidated target percentages are imposed by this PRD.

Goal	Candidate measure	How to interpret
Reduce manual effort	Time and number of steps to create RFQ, GRN, stage update, inspection and dispatch.	Compare against the customer's current method; measure task completion and correction rate.
Improve operational visibility	Share of open jobs with a valid stage update within an agreed interval; status-chasing calls/messages.	A low update rate may indicate UX or process friction, not simply employee non-compliance.
Improve material reliability	Material availability discrepancies; % receipts linked to PO/lot/certificate; shortage detection lead time.	Separate verified stock data from estimates and in-transit commitments.
Improve supplier follow-through	% supplier commitments with promised/revised date and follow-up owner; overdue commitment ageing.	Track whether the system surfaces risks before jobs are impacted.
Improve traceability	Time to locate a certificate; ability to trace job → lot → GRN → supplier → certificate.	Test with actual sample jobs, MTCs and lot records.
Improve quality control	% required inspections with completed evidence; open NCR age; repeat failure escalation.	Do not optimise pass rate by suppressing failures.
Improve maintenance visibility	% critical assets registered; downtime captured with affected jobs; PM due acknowledgement.	Separate recorded downtime from inferred availability.
Build adoption	Weekly active use by role; completed core workflows; data completeness; user-reported effort.	Measure whether EOSS becomes the source of operational coordination.
Demonstrate value	Founder-reported hours saved; missed follow-ups prevented; avoidable rework/status chasing reduced.	Financial impact is reported only when supported by records and validated with the customer.

4.1 Product outcome

OUTCOME STATEMENT

A founder should spend less time collecting updates, reconciling information and chasing routine follow-ups because EOSS captures work at source, connects the records and prepares the next action.

5. Product scope and release definitions

5.1 Three different scope layers

Layer	Purpose	Coverage	Non-goal
Strategic EOSS platform vision	Longer-term modular Business OS for small businesses across operating models.	Shared parties, work, transactions, documents, approvals, messages, sites and intelligence; future industry packs.	Does not mean all cross-industry functions ship in the manufacturing pilot.
Manufacturing-first product MVP	First real customer value in Indian manufacturing/fabrication.	Connected sales/order, suppliers/procurement, inventory/stores, jobs/production, quality, assets/maintenance, documents, dispatch and operational intelligence.	Not a complete accounting, payroll, statutory or advanced MRP suite.
Full-coverage interactive prototype	Let the product owner and pilot stakeholders experience the entire proposed manufacturing product before committing to production architecture.	12 workspaces, 26 screens, all agreed sections/field groups, 89 entity coverage, 12 demo roles (corrected from 10) and interconnected mock data.	Not production security, real multi-tenancy, real accounting or live integrations.

5.2 Full-coverage prototype baseline

Dimension	Target baseline	Interpretation
Workspaces	12	All must be discoverable and represented; no Coming Soon screens.
Top-level screen identifiers	26	Mobile wizard variants may be sub-screens; each listed identifier has a route or a clearly mapped screen state.
Domain entities	89	Layered model from platform foundations through AI/import records.
Field coverage	Approximately 920	The Scope Matrix v1.1 claims coverage against Field Dictionary v0.2; exact v0.2 workbook needs reconciliation before schema freeze.
Reusable components	37	Shell, data display, forms, AI/approvals and feedback/navigation patterns.
Demo roles	10	Owner/GM, Sales, Buyer, Storekeeper, Production Supervisor, Operator, QC, Maintenance, Dispatch, Auditor.
Seed environment	2 sites, 8 store locations	Representative factory and warehouse context with realistic linked data.
Prototype persistence	Local browser persistence	Records and actions must survive hard reload; no production backend required.
AI behavior	Simulated by default	Label simulations; keep provider adapter replaceable. A live provider call requires separate approval.

5.3 Manufacturing pilot scope

Capability	Pilot depth	Required result
Company, sites, users and roles	Foundation	Company/site/store configuration; users, employees, roles, permission scope and approval policies.
Sales & customer orders	Light but connected	Customer, RFQ/manual enquiry, quote versions, accepted customer PO, contract/drawing attachments and order-to-job conversion.
Supplier management	Core	Supplier directory, multiple contacts/addresses, materials supplied, supplier documents, order and delivery history.
Procurement	Core / lightweight	PR, supplier RFQ, supplier quotation capture/comparison, supplier PO, promised delivery and partial receipt.
Inventory & stores	Deep / pilot-critical	Material master/specifications, site/store locations, opening stock, GRN, lot/heat/batch, QC hold, stock ledger, reserve/allocate, issue/return, transfer, adjustment, remnant/scrap.

Capability	Pilot depth	Required result
Job & production	Core	Jobs, configurable stages, ownership, dates, quantities, progress updates, blockers and evidence.
Quality & inspection	Core	Incoming/in-process/final inspection, checklists, measurements, pass/fail/hold, evidence, NCR, rework and release.
Assets & maintenance	Core / pilot-critical	Asset register, site/work-centre assignment, PM schedule, breakdown, downtime, spare consumption and documents.
Document Vault	Deep / pilot-critical	Upload, category/metadata, version, validity/expiry, entity links, search, access policy, reminders and history.
Dispatch & delivery	Light	Packing readiness, multi-job dispatch, partial quantities, carrier/LR, expected delivery, POD and delivery quantity.
Reports & AI	Foundational	Operational reports, exceptions, reminders, daily owner brief, certificate extraction assistance and contextual recommendations.
Security and governance	Foundational	Tenant/site scope model, RBAC, audit, encryption, private document access and AI permissions for production readiness.

5.4 Explicitly outside the first manufacturing pilot

- Full accounting ledger, tax calculation, GST portal filing, e-invoicing and e-way bill integration.
- Payroll processing, leave, attendance, PF/ESI, biometric or attendance-device processing.
- Advanced tender portal discovery/submission and autonomous bid submission. Lightweight enquiry capture remains in scope.
- Full CRM automation and multi-channel inbox orchestration.
- Advanced MRP, finite-capacity scheduling, multi-level production planning, shop-floor IoT, machine telemetry and predictive maintenance.
- Advanced costing/valuation, landed-cost accounting, depreciation and statutory fixed-asset accounting.
- Advanced WMS optimisation, wave picking, replenishment algorithms and complex barcode automation. Basic location/bin tracking may be configured.
- Complex offline write synchronisation and automatic replay of consequential actions. Cached read-only access and saved drafts may be considered; actions must not auto-execute on reconnect.
- Live external integrations and real AI provider calls in the current low-cost prototype unless explicitly approved.

5.5 Broader Enterprise OS capabilities to preserve for roadmap reconciliation

The older cross-industry Enterprise OS BRD and UX Blueprint v0.3 contain additional capabilities that should not be silently discarded, but they are not included in the current manufacturing Scope Matrix v1.1. Retain them as strategic backlog candidates pending a separate product-owner prioritisation decision:

- Unified customer inbox, Gmail/Outlook RFQ intake and email thread digest/follow-up.
- WhatsApp inbound document/update intake, staff status capture, user-approved outbound messages and voice-first interactions.
- Collections sprint, overdue receivables, disputes-to-tasks, delivered-not-billed and finance/ledger integrations.
- Tender intake, eligibility evidence matching, bid pricing draft and approved bid-pack export.
- People directory, attendance/leave/OT, contract labour, expenses and payroll depth.
- Cross-industry service, project, trading, logistics and other configurable industry packs.

These broader capabilities are in the cross-industry BRD/UX source set (P0/P1/P2 classification varies by feature). They are not reclassified as manufacturing MVP commitments by this document.

6. Information architecture and screen inventory

6.1 Primary navigation

#	Workspace	Purpose
1	Home / Business Pulse	Executive awareness, exceptions, daily brief, quick access and approval queue.
2	Sales & Orders	RFQ, quotation versions, customer PO, amendments and job creation.
3	Suppliers	Supplier profile, GST registrations, materials supplied, orders, documents and delivery history.
4	Procurement	PR, supplier RFQ, supplier quotation comparison, supplier PO, commitments, receipts and returns.
5	Inventory & Stores	Material catalogue, stock overview, locations, lots, ledger, allocation, issue, transfer, count and remnant.
6	Jobs & Production	Job board, job workspace, stages, updates, material/procurement links and blockers.
7	Quality	Inspection queue, checklists, inspection records, NCR/rework and QualityRelease.
8	Assets & Maintenance	Asset register, PM calendar, breakdown reports, maintenance events and spares.
9	Dispatch	Dispatch planning, multi-job lines, lot links, document packs, delivery confirmation and POD.
10	Document Vault	Central documents, version history, expiry/obligations, classification and entity links.
11	Reports & Insights	On-time delivery, material consumption, supplier performance, QC, downtime and yield.
12	Admin & Security	Company, sites, stores, users, roles, permissions, approval policy, AI settings and audit.

6.2 Screen inventory — 26 top-level identifiers

ID	Screen	Required contents	Primary users
UX-01	Home / Business Pulse	Exceptions by category, daily brief, recent activity, quick actions, approval queue.	Owner / GM
UX-02	RFQ & Quotation	RFQ list/detail/lines; quote versions; active quote; version comparison; convert/mark lost.	Sales / Owner
UX-02b	Customer PO & Job Creation	PO header and lines, quote comparison, drawing/revision, delivery schedule, multi-line job creation.	Sales / Owner / Planner
UX-amend	Customer PO Amendment	Amendment record, changed-field diff, affected-job impact and review.	Sales / Owner
UX-03	Supplier Workspace	Supplier list and detail: profile, GST, materials, open orders, delivery history, documents, activity.	Buyer / Owner
UX-04	PR / Supplier RFQ / Quote Comparison	PR list/detail, supplier RFQ, side-by-side supplier quote comparison and recommendation.	Buyer
UX-05	Supplier PO & Receipt	PO list/detail, lines, commitments, amendments, overdue flags and GRN links.	Buyer / Storekeeper
UX-05b	Supplier Return	Return record, lot/qty, reason, NCR link and outward challan.	Buyer / Storekeeper
UX-05s	GRN / Goods Receipt	PO-linked receipt, multi-line quantities, lot/heat, certificate upload, store confirmation; mobile wizard.	Storekeeper / QC
UX-06	Inventory Overview	Stock on-hand/available/reserved/QC hold/in-transit, shortage flags, category/store/UOM filters.	Storekeeper / Owner
UX-07	Material Lot Detail	Lot/specs, dual-UOM, certificate, allocations, append-only movement ledger and lineage.	Storekeeper / QC
UX-cert	Certificate Extraction	Original document, extracted values, confidence, confirm/edit/dismiss and AgentRun provenance.	Storekeeper / QC
UX-transfer	Stock Transfer	From/to store, lots/qty, pending transfer and receipt confirmation.	Storekeeper
UX-cycle	Cycle Count	Expected vs counted qty, variance and adjustment approval.	Storekeeper
UX-remnants	Remnants & Scrap	Child lot, parent reference, disposition and scrap record.	Storekeeper

ID	Screen	Required contents	Primary users
UX-08	Job Board	Filterable jobs by site/customer/stage/priority/due date/blocker/material readiness.	Supervisor / Owner / Planner
UX-09	Job Workspace	Logical areas: Overview, Materials, Procurement, Production, Quality, Assets, Documents, Dispatch, Activity.	Supervisor / Owner / All
UX-10	Stage Update	Desktop inline update; mobile stage/quantity/action/evidence flow; draft/pending-sync state.	Supervisor / Operator
UX-11	Quality Inspection	Incoming/in-process/final queue, checklist, numeric tolerance, evidence, NCR and release.	QC Inspector
UX-ncr	NCR / Rework Workspace	NCR list/detail, disposition, rework order, reinspection and second-fail escalation.	QC / Supervisor
UX-12	Assets & Maintenance	Asset register/detail, PM calendar, event history, meter readings, mobile breakdown report.	Maintenance Tech / Supervisor
UX-dispatch	Dispatch	Multi-job/multi-PO lines, lot links, document pack, finished-goods stock movement and POD.	Dispatch / Sales
UX-13	Document Vault	Category filter, document list/detail, preview, versions, entity links, AI fields and expiry obligations.	Admin / All
UX-reports	Reports & Insights	Six core operational reports with date-range, table and chart views.	Owner / GM
UX-14	Admin & Security	Company, sites/stores, users/memberships, roles/grants, approvals, AI settings and audit summary.	Admin
UX-15	Audit & Activity Timeline	Append-only event log, filters, before/after detail and auditor read-only view.	Auditor / Admin

6.3 Job Workspace tab structure

The canonical product blueprint defines nine logical areas. The current prototype implementation has decomposed some areas into separate tabs, resulting in eleven visible tabs: Overview, Stages, Materials, Procurement, Quality, NCR/Rework, Assets, Documents, Dispatch, Updates and Blockers. This is acceptable provided all nine logical areas remain present and linked.

Blueprint logical area	Current prototype presentation
Overview	Overview
Materials	Materials
Procurement	Procurement
Production	Stages + Updates
Quality	Quality + NCR/Rework
Assets	Assets
Documents	Documents
Dispatch	Dispatch
Activity	Updates + Blockers

6.4 Navigation and global interaction

- Desktop: persistent left navigation with all workspaces discoverable, top bar with site context, notifications, active demo user/role and global quick access.
- Mobile: task-first navigation; compact bottom navigation with primary workspaces and a full workspace list under More. Use task-specific mobile flows rather than shrinking desktop tables.
- Provide global record search or command entry as a progressive enhancement; users must still be able to navigate and complete tasks without chat.
- Every record reference should support deep navigation to its detail view, preserving the user’s context where possible.
- Filters, tabs, menus, buttons and row actions must be functional. No dead links, decorative controls or toast-only success for a state-changing action.

7. Functional requirements by workspace

The following requirements describe the manufacturing-first MVP capability. Field-level detail is maintained in the companion field dictionary and scope matrix; this PRD defines expected product behaviour and acceptance.

7.1 Home / Business Pulse

- FR-HOME: Present a concise owner view of operational situations, not a wall of generic KPI tiles. Prioritise material shortages, supplier delays, blocked jobs, machine downtime, quality holds and expiring/expired documents.
- FR-HOME: Surface a focused 3–5 prioritised situation cards by default, with grouped overflow for related lower-priority situations. Never silently suppress a critical situation. This presentation rule is drawn from the broader Pulse UX model and should be applied without hiding manufacturing exceptions.
- FR-HOME: Each exception states what happened, why it matters, impacted records, freshness/source and the next useful action. Link directly to the related job, PO, lot, asset, inspection or document.
- FR-HOME: Show operational summary values derived from the shared data store. Do not duplicate counters or show counts that disagree with workspace records.
- FR-HOME: Provide quick access to open purchase requests, pending QC inspections, active jobs, approvals and other common tasks.
- FR-HOME: Provide a contextual Ask EOSS / Companion entry as an optional way to investigate or prepare actions. The normal workspace UI remains fully available.

7.2 Sales & Orders

- FR-SALES: Create and manage customer RFQs/enquiries with customer, received date/channel, required-by date, line descriptions, quantities, UOM, material specs, drawing number/revision and source attachments.
- FR-SALES: Flag incomplete RFQs and missing drawing references; do not infer missing specifications. Preserve clarifications and source evidence.
- FR-SALES: Create versioned quotations with validity, line quantities/UOM, rates, discounts, delivery/payment terms, lead time, currency, notes and attachments. Issued versions remain immutable; revisions create a new version.
- FR-SALES: Record customer PO/contract and compare it with the accepted quotation. Highlight changes in quantity, rate, specification, drawing revision, delivery or terms.
- FR-SALES: Support customer PO amendments with a before/after diff and impact review for linked jobs, allocations and open quantities.
- FR-SALES: Allow one PO to create multiple jobs and a job to fulfil a partial order quantity; maintain remaining quantities.

7.3 Suppliers

- FR-SUP: Maintain customer/supplier party records with legal and trading names, classification, contacts, addresses, GST registrations where applicable, payment terms, active status and notes.
- FR-SUP: Show supplied materials, preferred/candidate status where supported, open supplier POs, commitments, receipt/delivery history, quality documents and related activity.
- FR-SUP: Allow multi-contact, multi-address and multi-state GST registration records. Protect sensitive fields according to role permissions.
- FR-SUP: Provide a clear supplier summary so a buyer can understand current commitments and delivery reliability without searching across workspaces.

7.4 Procurement

- FR-PROC: Create purchase requisitions from job material requirements, maintenance needs or minimum-stock requirements. Capture required date, target site/store, reason, requester, priority and quantity/UOM.
- FR-PROC: Create supplier RFQs and capture supplier quotations including validity, offered rate, currency, MOQ, lead time, freight/tax/discount and delivery promise.
- FR-PROC: Compare offers on a normalised basis using compatible UOM, quantity, taxes/freight and delivery terms. Show missing or non-comparable values explicitly.
- FR-PROC: Create and submit supplier POs, capture promised dates and log commitment follow-ups with owner, revised promise, outcome and notes.
- FR-PROC: Support partial receipt and outstanding quantity. A Supplier PO is a commitment, not stock.
- FR-PROC: Support supplier PO amendment and short-close with reason, supplier acknowledgement and impact on affected jobs.

- FR-PROC: Create supplier return records from rejected/held receipt disposition with lot, quantity, reason, NCR reference and challan details.

7.5 Inventory & Stores

- FR-INV: Maintain material masters with category, grade, dimensions, attributes, primary/secondary UOM, tracking mode, reorder point and conversion method.
- FR-INV: Show stock positions by site, store and item: on-hand, reserved, QC hold, blocked, in-transit and available. Available is derived, not manually overwritten.
- FR-INV: Support the eight configured StoreLocations in the prototype, filters by location/category/UOM and location-level item counts.
- FR-INV: Support lot/heat/batch tracking for configured items. Non-lot-tracked consumables remain aggregate stock and must not be represented by fake lot records.
- FR-INV: Support GRN, issue, return, transfer, adjustment, cycle count, remnant and scrap flows as enabled by release scope.
- FR-INV: Every stock-changing activity creates a movement with quantity, UOM, timestamp, source/destination, reference, actor and reason. Corrections use reversal/correction entries, not silent edit-in-place.
- FR-INV: Dual-UOM values must be item-specific and explainable. Support direct factor, weight-per-area, weight-per-length or explicit formula/parameter approaches; do not use one universal kg-to-length conversion.
- FR-INV: Lot detail must expose supplier, GRN, heat, grade, dimensions, certificate, allocations, movements and parent/child lineage.

7.6 Jobs & Production

- FR-JOB: Create a job from one or more customer PO lines or through an explicitly authorised non-PO route where the pilot customer confirms the use case. Capture authorisation type and reason for non-PO jobs.
- FR-JOB: Use configurable stage templates and snapshot the selected template/version when the job is created. Assign site, owner, priority, planned dates, work centre and relevant asset.
- FR-JOB: Provide a filterable job board and a connected Job Workspace with nine logical capability areas. The current 11-tab implementation may split areas for clarity.
- FR-JOB: Create job material requirements and compare needed quantities with available/reserved/in-transit stock. Show material readiness and affected supplier commitments.
- FR-JOB: Record stage start/hold/complete, actual dates, completed_good_qty, rejected_qty, rework_qty, scrap_qty, notes, photos/evidence and blocker.
- FR-JOB: Validate quantity balance at stage completion: input_qty must equal completed_good_qty + rejected_qty + rework_qty + scrap_qty, unless a reasoned exception is authorised.
- FR-JOB: Capture performed_by_employee_id separately from logged_by_user_id. The employee may not have an EOSS user account.
- FR-JOB: Allow any permitted team member to raise a blocker. Notify the job owner and relevant operational workspace based on blocker type.

7.7 Quality

- FR-QA: Support incoming, in-process and final inspections using configurable inspection templates and checks.
- FR-QA: Capture sample quantity, accepted/rejected quantity, pass/fail/hold, numeric values, tolerance deviations, notes, photos and inspector.
- FR-QA: Create NCRs from failed inspections or non-conforming material/stage events. Link source record, lot/job/stage, affected quantity, severity, containment, disposition and owner.
- FR-QA: Support rework order, reinspection and escalation on repeated failure.
- FR-QA: Create QualityRelease only when required checks and certificate dossier are complete and open NCRs have acceptable disposition, unless an authorised exception is recorded.
- FR-QA: Block dispatch when required final quality release is missing.

7.8 Assets & Maintenance

- FR-ASSET: Maintain asset register with asset code, site, work centre, category, criticality, manufacturer/model/serial, meter type and operational status.
- FR-ASSET: Support preventive maintenance by calendar or meter reading; due occurrences and service history are visible.
- FR-ASSET: Provide mobile breakdown reporting: asset, severity, symptoms, occurrence/downtime timestamps, evidence and affected jobs/stages.
- FR-ASSET: Create maintenance event/work order with technician, diagnosis, work performed, return-to-service test, approval where required and downtime end.

- FR-ASSET: Link affected jobs through MaintenanceEventJobLink rather than an array embedded in the event.
- FR-ASSET: Consume spare parts through a referenced stock movement and link maintenance spare consumption to the event.

7.9 Dispatch & Delivery

- FR-DSP: Create dispatches that can contain multiple jobs and multiple customer PO lines.
- FR-DSP: Capture carrier, vehicle, LR number, dispatch date, expected delivery, dispatch lines, quantity/UOM, lot links and destination.
- FR-DSP: Validate dispatch quantity against remaining customer order/job quantity and required QualityRelease.
- FR-DSP: Create a document pack checklist covering relevant certificates, inspection reports, delivery challan and other required evidence.
- FR-DSP: On confirmed dispatch, create the corresponding finished-goods stock-out movement and update dispatch status.
- FR-DSP: Capture delivery confirmation, actual delivery date, received quantity, POD photo/document and discrepancy/short-delivery quantity.
- FR-DSP: Partial dispatch/delivery must leave the unfulfilled balance visible.

7.10 Document Vault

- FR-DOC: Provide one central document repository with type/category, title, owner, site, sensitivity, issuing party, number, version, issue/validity/expiry dates, tags and retention metadata where applicable.
- FR-DOC: Link a document to one or more canonical records: company/site, supplier, material lot, asset, job, inspection, dispatch or other supported entity.
- FR-DOC: Preserve the original file and document version. A new version does not silently overwrite prior evidence.
- FR-DOC: Surface expiry and renewal obligations with responsible person and reminder horizon.
- FR-DOC: Support preview, search/filter, upload from desktop and camera/file attachment from mobile. AI classification or link suggestions remain proposals until user confirmation.
- FR-DOC: Apply document-level access control. Each restricted download is logged; a UI-hidden button alone is not security.

7.11 Reports & Insights

- FR-REPORT: Provide operational reports for on-time delivery, material consumption by job, supplier performance, QC pass rate, maintenance downtime and job yield.
- FR-REPORT: Allow date range and appropriate site/customer/supplier/job filters. Provide tabular and chart views where meaningful.
- FR-REPORT: Ensure report KPIs reconcile with source workspace records and the shared business clock.
- FR-REPORT: Show metric definitions, source data freshness and limitations where calculations are approximate.
- FR-REPORT: Report export is not granted by default to the Auditor; it may be explicitly granted and scoped. Bulk record export remains denied unless explicitly approved.

7.12 Admin, Security & Audit

- FR-ADMIN: Manage company profile, legal entity, sites and store locations, users, memberships, employees, roles, grants and approval policies.
- FR-ADMIN: Provide prototype role switcher for demonstration. The switcher is not production authentication or security enforcement.
- FR-ADMIN: Provide AI settings with Full, Limited and Paused modes and a visible usage/cost control for authorised administrators.
- FR-ADMIN: Record audit events for consequential actions, approvals, role changes, document downloads and material/dispatch/QC state changes.
- FR-ADMIN: Provide Audit & Activity Timeline with actor, action, record reference, time and before/after values, subject to scope.
- FR-ADMIN: Production permissions must be enforced server-side, deny by default, and evaluate tenant/site/record/field/document scope.

8. End-to-end workflows and business rules

8.1 Customer demand to job

Stage	Required behaviour
1. Enquiry / RFQ	Create customer RFQ and lines, capture channel/date, quantities, UOM, requirements, drawing revisions and source attachments. Flag missing technical information; do not infer specs.
2. Evaluate and quote	Capture clarifications and technical feasibility; create quote versions with immutable issued snapshots, delivery/payment terms and approval where margin/discount policy requires.
3. Customer PO	Record accepted quote version, customer PO/contract, line quantities, dates, terms and drawing revision. Compare against quote and highlight changes.
4. Job creation	Create one or more jobs from PO lines; preserve PO/quote/drawing lineage, planned dates, owner, site, priority and stage-template snapshot.

8.2 Material planning and procurement

- Create JobMaterialRequirement from job demand or an approved requirement source. Show on-hand, reserved, available, QC-hold and in-transit quantities before declaring shortage.
- Create PR and supplier RFQ for shortage/min-stock/maintenance demand. Supplier quotes are compared on a like-for-like basis; expired offers cannot be selected without an explicit override.
- Create Supplier PO only after required approval. PO commitment is not receipt and does not increase stock.
- Track promised date per line/schedule; log revised dates, supplier contact, reason, follow-up owner and status. Overdue supplier commitments update affected job risk.

8.3 GRN, lot creation and incoming quality

- Receive against Supplier PO where possible; non-PO receipt is a controlled exception requiring reason and optional retro-PO reference.
- Record received, accepted, rejected and hold quantities separately. Validate against outstanding PO balance and receipt tolerance; out-of-tolerance quantity requires explanation/approval.
- Only accepted quantity becomes usable stock. Rejected quantity is segregated to quarantine; hold quantity remains blocked until disposition.
- Create MaterialLot for lot/heat-tracked items; do not create MaterialLot records for non-lot-tracked consumables or unreceived supplier commitments.
- Preserve MTC/COA original. Certificate values are manually entered or AI-proposed; a human verifies values and links the certificate to the correct lot(s).

8.4 Inventory availability and movement

> **INVENTORY INVARIANT**
>
> Available quantity is derived from posted stock movements and valid reservations/holds. It is not a manually editable balance.

- On-hand quantity is the net sum of posted stock movements for the relevant item/specification, lot and location context.
- Available quantity is computed from on-hand less active reservations, QC holds and other blocked quantities. In-transit quantities are shown separately and are not counted as available at the destination.
- Allocation reserves stock against a JobMaterialRequirement. Prevent over-allocation unless an authorised override with reason is recorded.
- Issue/return records source lot, destination, quantity/UOM, actor, receiver, transaction time and reference. Return should reference original issue where practical.
- Transfer is represented as dispatch/transfer-out and receipt/transfer-in events. Destination stock is not available until received.
- Adjustment and correction are auditable. Do not edit posted movement history; use reversal/correction movements.
- Remnant creation creates a child lot with parent/source link, dimensions, quantity, condition and disposition. Scrap is never treated as usable stock.

8.5 Production stage execution

- Each JobStage references the job, sequence/template, work centre and optionally an asset, owner and planned/actual dates.
- Stage updates are append-oriented events with action, quantities, note/evidence, transaction time, performer employee and recording user.
- completed_good_qty is the standard field name for good output. Do not mix it with a generic completed_qty field.
- On completion, validate input_qty = completed_good_qty + rejected_qty + rework_qty + scrap_qty. Require an explanation/authorised exception for unaccounted quantity.
- Machine, material, manpower, customer, design or other blockers link to the job and notify the relevant owner/workspace.

8.6 Quality, NCR and release

- Inspection result may be Pass, Fail or Hold; failed or held material/work remains unavailable for downstream use until disposition.
- NCR links to the source inspection and affected lot/job/stage. Disposition may include rework, return, scrap or approved deviation, subject to permissions.
- Rework order links to NCR and reinspection. Repeated failure escalates to owner/authorised role.
- Final QualityRelease requires completion of mandatory inspections and certificate dossier. Dispatch readiness requires QualityRelease unless an approved, reasoned exception exists.

8.7 Maintenance and dispatch

- Breakdown reports link the asset to affected jobs/stages via MaintenanceEventJobLink. The event records downtime, diagnosis, work, spares and return-to-service checks.
- Dispatch may include multiple jobs and PO lines. DispatchLotLink connects each dispatch line to lots and quantities; do not store lot relationships as JSON arrays.
- Confirmed dispatch posts finished-goods stock-out movement. POD/delivery confirmation records actual received quantity, evidence and discrepancies.
- InvoiceStatus and payment status are reference/status records only in the manufacturing MVP; EOSS does not become a full accounting ledger.

9. Data model and entity catalogue

9.1 Data model principles

- Use stable IDs and canonical relationships across workspaces. Related screens read the same record rather than maintaining duplicated local copies.
- Keep entity definitions, view models, seed fixtures, services and UI presentation separate. In the prototype, the mock service layer can be replaced by production APIs later.
- Use explicit relationship/link entities for many-to-many relationships where lineage, quantity or provenance is important.
- Keep stock ledger and audit history append-only. Separate transaction_at (when the physical event occurred) from posted_at (when recorded/posted) where supported.
- Preserve source lineage across RFQ → quotation → customer PO → job; supplier RFQ → quotation → supplier PO → GRN → lot → certificate → allocation → job; and job → QC release → dispatch → POD.
- Use item-specific UOM conversion rules and parameters. Avoid a universal conversion scalar for metals where weight/area/length depends on grade, dimensions, profile or thickness.

9.2 Entity dependency layers — 89 entities

Layer 0 — Platform / foundation (15)

Tenant, Company, LegalEntity, Site, StoreLocation, User, Membership, Role, RoleGrant, ApprovalPolicy, ApprovalRequest, AuditEvent, Notification, UOM, MaterialCategory

Layer 1 — Master data (18)

Party, PartyContact, PartyAddress, PartyGSTRegistration, MaterialItem, MaterialAttributeDefinition, SupplierMaterial, Asset, WorkCentre, Document, DocumentVersion, DocumentLink, DocumentObligation, StageTemplate, StageTemplateStep, InspectionTemplate, InspectionTemplateCheck, Employee

Layer 2 — Commercial transactions (16)

CustomerRFQ, RFQLine, CustomerQuotation, CustomerQuotationLine, CustomerPO, CustomerPOLine, CustomerPOAmendment, SupplierRFQ, SupplierRFQLine, SupplierQuotation, SupplierQuotationLine, SupplierPO, SupplierPOLine, SupplierPOAmendment, SupplierCommitment, PurchaseRequisition

Layer 3 — Job and production (7)

Job, JobLine, JobMaterialRequirement, PurchaseRequisitionLine, JobStage, JobUpdate, JobBlocker

Layer 4 — Inventory transactions (9)

GoodsReceipt, GoodsReceiptLine, MaterialLot, StockMovement, MaterialAllocation, StockTransfer, StockTransferLine, StockCount, StockCountLine

Layer 5 — Quality and dispatch (10)

MaterialCertificate, CertificateLotLink, Inspection, InspectionResult, NCR, ReworkOrder, QualityRelease, Dispatch, DispatchLine, DispatchLotLink

Layer 6 — Maintenance and dispatch completion (8)

MaintenancePlan, MaintenanceOccurrence, MaintenanceEvent, MaintenanceEventJobLink, MaintenanceSpareConsumption, AssetMeterReading, DeliveryConfirmation, InvoiceStatus

Layer 7 — AI, import and supplementary (6)

AgentRun, AgentActionProposal, ImportJob, SupplierReturn, PaymentStatusEvent, DispatchDocumentPack

9.3 Important relationship and schema corrections

Area	Required model rule
Party GST	PartyGSTRegistration stores multiple GST registrations per party/state and validity period.
Supplier PO revisions	SupplierPOAmendment records changes to supplier PO quantity/date/price/other terms with acknowledgement and status.
Dispatch lot lineage	DispatchLotLink links DispatchLine to MaterialLot with lot quantity and UOM; no lot_ids[] JSON on DispatchLine.

Area	Required model rule
Maintenance impact	MaintenanceEventJobLink links a maintenance event to affected Job/JobStage; no affected_job_ids[] JSON on MaintenanceEvent.
Shop-floor identity	JobUpdate stores performed_by_employee_id and logged_by_user_id separately; an Employee may exist without a User login.
Stage quantities	Use completed_good_qty consistently, together with rejected_qty, rework_qty and scrap_qty.
UOM conversion	Use conversion_method plus conversion_params_json; include direct factor, weight-per-area, weight-per-length and explicit formula parameters as required by item type.
Stock movement	StockMovement is append-only. Availability is derived from ledger and reservations/holds. Track transaction_at separately from posted_at.
Material ownership	owner_party_id on lot is conditional on customer-owned material being confirmed for the pilot.
Documents	Document identity and version are separate; expiry/validity belongs to the appropriate document version and obligation model, not duplicated conflicting fields.

9.4 Field dictionary status

The current Scope Matrix v1.1 states that every entity from Field Dictionary v0.2 is represented in a form, detail view, system-field section, expandable section or table column. It reports approximately 920 fields. The v0.2 workbook was not present in the source bundle assembled for this PRD; a v0.1 workbook was present. Treat the v1.1 entity/field coverage statement as the current planning baseline, but do not freeze a production schema until the exact v0.2 workbook and accepted deltas are reconciled.

10. Roles, permissions and approvals

10.1 Ten demo roles

Role	Primary permissions / responsibilities
Owner / GM	Business-wide operational view; approve consequential actions; configure company, sites, access, AI and approval policies.
Sales	Manage RFQs, quotations, customer POs, commercial documents and own customer delivery visibility.
Buyer / Procurement	Manage PR, supplier RFQ/quotes, supplier POs, supplier commitments and returns.
Storekeeper	Receive GRNs, create lots, upload certificates, issue/return/transfer stock and count inventory in assigned stores.
Production Supervisor	Manage assigned jobs, stages, blockers, material readiness, PR initiation and production coordination.
Operator	View assigned jobs/stages, record stage completion and evidence, confirm permitted issues; no approve/export.
QC Inspector	Perform incoming/in-process/final inspection, upload evidence, raise NCR and execute permitted release actions.
Maintenance Technician	Report breakdown, manage maintenance events, record downtime, consume spares and record return-to-service.
Dispatch Personnel	Prepare dispatch, manage dispatch documents, confirm dispatch and record POD/delivery.
Auditor	Scoped read-only access; permitted document downloads are individually logged; no create/edit/approve/bulk export.

10.2 Permission model

Permission evaluation must combine tenant scope, site scope, role grant, record ownership/assignment, action type and field/document classification. Deny by default. Separate view, create, edit, submit, approve, reject, delete, export and download actions; approval is not equivalent to edit permission.

- Role and site grants are explicit. Users may access one or multiple sites/stores. Cross-site views require an explicit grant.
- Operator cannot approve or export. Auditor is read-only; individual document downloads may be allowed and logged. Bulk record export and audit export are denied by default.
- Rejecting an approval requires a reason. Approval captures requester, approver, decision, timestamp, reason and an immutable action snapshot.
- Maker-checker applies to defined high-impact actions: high-value POs, stock adjustment/negative stock, substitutions, scrap, QC release after hold/fail, return to service after critical breakdown, role/security changes and external sends.
- Prototype RoleSwitcher only demonstrates role-specific visibility and action affordances. It is not a security boundary.

10.3 Production security release gate

NO REAL CUSTOMER DATA BEFORE THIS GATE

Tenant isolation, server-side RBAC, secure document storage/access, encryption, audit and tested permission boundaries must be implemented and verified before a production pilot. Client-side hiding of controls is not enforcement.

11. AI, intelligence and automation

11.1 AI is an intelligence layer

AI should be present where it reduces effort, not as a separate mandatory navigation model. Users must always be able to complete core workflows through ordinary screens and structured forms. Pulse is the home awareness surface; a contextual Companion / Ask EOSS conversation may open from a situation card or record when the user wants to investigate or prepare an action. It is optional and does not replace structured forms for quantities, inspections, approvals or configuration.

AI / automation capability	Mechanism	MVP treatment
Business Pulse exceptions	Deterministic rules over job, stage, inventory, supplier, QC, asset and document records.	Core. Every exception links to source records and is recalculated after relevant state changes.
Daily owner brief	Rules gather due/overdue situations; AI may explain in plain language.	Simulated in prototype; source-linked and timestamped.
Job risk flag	Rules detect overdue stage, material shortage, supplier promise risk, machine downtime or QC hold.	Core rules-based. No predictive ML claim without history.
Material shortage/matching	Deterministic match by material, specification, grade, dimensions, UOM, site/store, available quantity and condition.	Core. Matching criteria visible; substitutions require human approval.
Supplier follow-up	Reminder based on response due/promised/revised date and follow-up history.	Rules-based. User-controlled cadence and ownership.
Certificate extraction	Extract supplier, certificate number, heat/lot, grade, dimensions, test values and issue date.	Assistive prototype simulation by default. Human verifies each field; preserve original file and provenance.
Document classification/link suggestion	Suggest document type/category and candidate entity links.	Simulated suggestion; user confirms.
Production update summarisation	Turn short note/voice transcript into suggested stage note or blocker category.	Future/P1 candidate; user confirms before writing.
Operational Q&A / Ask EOSS	Read-only answers over accessible structured records, with links and data freshness.	Simulated prototype capability; permissions must be respected in real implementation.
RFQ extraction	Extract customer, items/specs, quantities, due date and missing information from email/document.	Prototype/demo candidate; live inbox integration is future scope.

11.2 AI action lifecycle

1. Understand: identify the user, tenant/site, intent, target record and ambiguity. Ask for clarification when identifiers are ambiguous.
2. Retrieve: access only records permitted to the caller; include source, last-updated time, status and missing facts.
3. Plan: show proposed changes, affected records, quantity/date/money effects, risks and approvals.
4. Approve: apply RBAC, maker-checker and approval policy. Consequential actions require explicit confirmation.
5. Execute: use controlled service adapters, idempotency key and transaction boundaries. Do not duplicate stock movements, orders or external sends.
6. Prove: report success/failure only after application/provider confirmation; link resulting record and audit event.

11.3 Human-control and provenance requirements

- AI-extracted values must not be saved to canonical records without explicit review/confirmation.
- Use qualitative confidence (High/Medium/Low) for non-technical users; store model/version/source metadata separately.
- Every insight must cite the source record(s) and data freshness. Never generate a status, metric or financial amount not supported by records.
- AI must not access records outside tenant/site/role/document scope. It must not be a permission bypass.
- When AI is paused or unavailable, all normal non-AI workflows remain usable.
- Provide tenant-level AI modes Full, Limited and Paused; record agent runs and allow an emergency kill switch in production.

12. UX, visual identity and interaction requirements

12.1 Experience north star

> **SIMPLICITY WITH DEPTH**
>
> The product should feel sophisticated underneath and remarkably simple on the surface. More capability, less weight, less friction and less effort.

12.2 Visual direction under active review

The latest product-owner feedback favours a visually rich, dark-first Industrial Intelligence / manufacturing command-centre direction inspired by a supplied dark manufacturing dashboard reference. This is a direction to explore, not an approved final design system. It must be tested in the real application before being propagated across all workspaces.

Design dimension	Target direction	Guardrail
Colour / surfaces	Deep charcoal and navy foundations; restrained cyan/teal for intelligence and activity; steel blue for structure; amber/red for meaningful warnings.	Avoid neon saturation and excessive glowing effects. Preserve long-session readability.
Typography	Strong, legible type hierarchy; clear numeric/data typography; readable small labels; monospace only for record IDs/codes where useful.	Do not reduce readability to fit more information.
Composition	Distinctive workspace-specific layouts, operational visualisations, meaningful panels and balanced space.	Do not repeat the same KPI-card-plus-table layout everywhere.
Data visualisation	Production flow, material distribution, job dependencies, time/status, trend and traceability visuals where data supports them.	No arbitrary charts, fake OEE or incompatible-unit aggregations.
Interaction	Contextual actions, progressive forms, drawers, filters, saved views, quick capture and responsive task flows.	Visual richness must not add steps or hide capabilities.
Motion / effects	Purposeful transitions and clear interaction feedback.	No animation for decoration; respect reduced-motion preferences.

12.3 Three reference experiences

Business Pulse — executive intelligence centre

- Show operational health across production, materials, quality, assets and dispatch.
- Prioritise actionable situations with impact, affected records, responsible owner, deadline/freshness and next action.
- Use rich visual hierarchy and meaningful operational flow; avoid redundant exception counters and oversized KPI tiles.
- Show Needs Your Decision only for real, resolvable approval/action records; never display undefined labels.

Inventory — visual inventory intelligence

- Show stock distribution across all configured locations, item/lot counts, shortage and QC hold signals, with filters and detailed list views.
- Use comparison measures that are valid. Do not sum kg, pieces, metres and cylinders into a meaningless total.
- Provide clear available/reserved/QC hold/in-transit indicators and fast drill-down to lot, certificate, allocation and movement history.

Job Workspace — manufacturing control centre

- Present job identity, customer/PO/drawing, due date, owner, stage progress and job health in one coherent frame.
- Use the secondary space to show materials, blockers, procurement, quality, machine dependencies, documents and dispatch readiness; avoid unbalanced unused areas.
- Allow the user to see the whole job context without assembling it across disconnected screens.

12.4 Data-entry design standards

- Use progressive forms. Separate essential fields from optional/advanced fields; do not surface all ~920 fields in one form.
- Prefill from the current record and context. For example, a GRN started from a Supplier PO should prefill supplier, PO lines and outstanding quantities.
- Use searchable pickers and recent values for customer, supplier, material, lot, job, employee and location.
- Use inline editing where it is safe and clear; use structured forms for quantities, dimensions, inspections and controlled configuration.
- Use contextual validation at field or step level. State why a value is invalid and how to fix it.
- Use mobile task flows for GRN, stage update, inspection, breakdown, inventory lookup and POD. Keep touch targets large and typing minimal.
- Provide visible loading, empty, filtered, no-results, validation-error, permission-denied, pending-approval, success, provider-failure and stale/offline states.

12.5 Global interaction patterns

Search, filter, sort, status chip, record header, detail drawer, movement ledger, stage timeline, lot trace card, document card, exception card, activity feed, mobile cards, schema-driven forms, quantity inputs, entity pickers, inspection checklist, certificate uploader, AI proposal banner, approval request card, confirmation modal, wizard shell, contextual alert and notification dropdown should be standardised as reusable patterns.

13. Data capture, onboarding and migration

13.1 Assisted onboarding

1. Configure company, legal entity, sites, store locations, departments/work centres, users, employees, roles and approval rules with the pilot team.
2. Import customer and supplier directory, material catalogue/specifications, assets, documents and opening stock from CSV/Excel templates.
3. Preview import mapping, duplicates and validation errors before commit. Retain source file and import summary; do not silently overwrite existing records.
4. Load open customer POs and jobs, stage templates, current supplier POs/commitments, stock lots/certificates and relevant maintenance history.
5. Obtain owner sign-off on site/store masters, item UOM/conversion, opening stock and critical document links before operational use.

13.2 Capture-once rules

Information	Capture once in	Reuse in
Customer identity/contact/address	Party / PartyContact / PartyAddress	RFQ, quotation, customer PO, job, dispatch, delivery communication.
Supplier identity/GST/contact	Party and related party records	Supplier directory, RFQ, supplier quotation, PO, GRN, certificate, returns.
Material identity/spec/UOM	MaterialItem and attribute definitions	PR, supplier RFQ/PO, GRN, lot, allocation, issue, consumption and reports.
Job identity and commitment	Job / JobLine	Material requirements, stages, blockers, quality, documents, dispatch and Pulse.
Employee identity	Employee (optional User link)	Stage updates, maintenance work, issue/receipt activity and employee attribution.
Document identity/version	Document / DocumentVersion	Job, supplier, material lot, asset, inspection, dispatch and renewal obligation.

13.3 Adoption principles

Start with a narrow daily value loop and expand when data quality and usage are stable. A recommended sequence is open jobs and stage updates; material/store receipts and availability; supplier commitments and procurement; documents/certificates and QC; assets/maintenance; dispatch and reporting. Assisted onboarding and floor-level observation are part of the product rollout, not optional afterthoughts.

14. Integrations and platform boundaries

Integration / capability	Manufacturing MVP treatment	Boundary / future decision
CSV / Excel import	Required for assisted setup of parties, materials, assets, documents and opening stock.	Preview, validation, deduplication and import audit; do not assume full two-way sync.
Gmail / Outlook	Manual RFQ record and document attachment in first manufacturing pilot; email intake may be simulated.	Live inbox connection, thread parsing and sending requires separate scope and OAuth/security review.
WhatsApp Business	Not a hard dependency for the current manufacturing prototype.	Inbound docs/status and user-approved outbound messages are strategic candidates; provider policies/permissions must be validated.
Accounting systems (Tally / BUSY / Zoho Books)	Invoice/payment status reference only unless separately approved.	Define source of truth and sync direction before implementing. No silent accounting writes.
AI provider	Simulated AI by default in the full-coverage prototype.	Keep backend/provider adapter replaceable. Live call, key management, data processing terms and cost cap require explicit approval.
Barcode / QR	Optional accelerator, not a prerequisite.	Future phase after label format, device access and scan workflow validation.
Machine / IoT connectors	Not in first pilot. Manual breakdown/meter reading/maintenance capture.	Telemetry and predictive maintenance are future scope.

14.1 Prototype versus production boundary

Concern	Prototype requirement	Production requirement
Data store	Shared local mock-service data with deterministic seeds and browser persistence.	Authoritative transactional backend and database with migration strategy.
Authentication	Demo role/user switcher to exercise layouts and actions.	Managed identity, secure sessions, MFA for privileged users and revocation.
RBAC	UI-level simulation for demonstration only.	Server-side enforcement at API/storage boundary; deny by default.
Tenant/site isolation	Model and navigation demonstrate intended scopes; no production claim.	Enforced tenant and site isolation with automated negative tests.
Documents	Sample/local prototype assets and metadata.	Private object storage, ACL, encryption, malware scanning/retention decisions and audited downloads.
AI	Labelled simulated suggestions and deterministic rules.	Server-side adapters, policy enforcement, cost controls, run logs, provenance and kill switch.
Offline	Saved drafts / cached read-only behaviour where implemented; no complex conflict resolution.	Defined sync protocol, idempotency, conflict handling and safe recovery. Consequential actions never auto-execute.
Audit	Prototype activity simulation.	Immutable/auditable event store with actor, action, before/after, provenance and retention.

15. Non-functional requirements

Area	Requirement
Responsive	Support 390px mobile, 768/1024px tablet and 1366/1440px desktop. Core workflows usable at 200% zoom where feasible.
Accessibility	Keyboard navigation, visible focus, semantic headings/labels, screen-reader names for controls, accessible form errors and adequate contrast. Target WCAG 2.2 AA for user-facing workflows where feasible.
Performance	Proposed pilot target: p95 < 2 seconds for common list/detail reads and < 3 seconds for common writes under agreed pilot load. Benchmark with realistic dataset before claiming.
Reliability / idempotency	No duplicate stock postings, approvals, external sends or dispatch movements on retry. Use stable action IDs/idempotency where production adapters exist.
Data integrity	No silent edit-in-place of posted stock/issued quote/approved action. Corrections use versioning, reversal or amendment. Prevent unapproved negative stock or over-allocation.
Observability	Structured logs, metrics, traces/correlation IDs, retry/failed job visibility and integration health. Never log secrets or unmasked sensitive data.
Backup / recovery	Define RPO/RTO with engineering before production pilot and perform restore testing. Not claimed by the prototype.
Internationalisation	English initial UI; avoid hard-coded workflow text and support future localisation. Dates, units, numbers and currencies should be locale-aware.
Security / privacy	Encrypt data in transit and at rest in production; protect PII and commercial information; manage secrets securely; implement access, retention, export and deletion paths.
Low bandwidth	Prioritise task flows, visible loading/retry state and local drafts. Do not assume continuous connectivity on shop floor.

Performance and accessibility values above are proposed targets from the product definition and should be confirmed by engineering and pilot testing. Security/legal controls require specialist verification and do not constitute a compliance claim.

16. Prototype implementation requirements and current baseline

16.1 Reference implementation approach

The current prototype direction uses React 18, TypeScript 5, Vite, Tailwind/CSS tokens, Zustand, Dexie.js/IndexedDB, React Router, React Hook Form and Zod, with a typed mock-service layer and deterministic seed data. The PRD is tool-agnostic: code structure and service boundaries should be portable across Claude Code, Cursor, Grok Build, Emergent or other implementation agents.

16.2 Required separation of concerns

- Presentation: AppShell, navigation, workspace screens, reusable components and responsive patterns.
- Domain: typed entity and value definitions, business rules, status enums, quantities, relationships and validation.
- Services: mock/persistent service interface for reads, writes, transitions and derived views.
- State: shared application store; all workspaces read and update the same records.
- Seed fixtures: deterministic IDs and realistic relationships; include reset/reseed support for demos.
- AI adapter: simulated implementation by default; isolated service contract for future provider integration.

16.3 Current prototype status snapshot (reported, not independently re-tested for this PRD)

Area	Last reported state	Verification still required
Coverage	12 workspaces and 26 screen identifiers defined in Scope Matrix v1.1; initial screens exist across Home, Sales, Suppliers, Procurement, Inventory, Jobs, Quality, Assets, Dispatch, Documents, Reports and Admin.	Confirm every route/screen and detailed form/view is implemented and navigable.
Job Workspace	11 visible tabs reported, covering the nine logical blueprint areas.	Verify each tab contains linked data and working actions, not a shell.
Build	Last integration report stated a clean build with 2,109 modules and no TypeScript errors.	Re-run current build and test suite after visual work.
Store write-back	Create dispatch, raise NCR, close NCR and dispatch status transitions were reported wired to shared Zustand actions.	Verify all writes persist and update related records after browser reload.
Persistence	Last defect report stated Zustand state reset on hard reload. A later request asked for Dexie/IndexedDB or equivalent persistence.	Confirm create/update actions survive a hard refresh; do not mark persistence complete until tested.
Business Pulse	Last report noted seeded Pulse exceptions were static.	Confirm exception state reacts to resolved/changed records; counts and action links must reconcile.
Employee attribution	Employee seed and types were added; Stage Update employee selector was still noted as missing.	Verify performer employee and logged-in user are both captured and displayed.
Inventory locations	Eight locations were present in the seed according to the report; latest screenshots still visibly showed four location groups.	Verify all eight are exposed and the counts/filters reconcile.
Visual direction	A dark-first industrial command-centre direction has been requested; updated Pulse and Inventory screenshots were shared, but the user has not approved the design.	Treat dark-first art direction as under review. Business Pulse, Inventory and Job Workspace remain reference screens for visual iteration.

16.4 Prototype acceptance principles

- Every visible primary control works; no toast-only state-changing actions.
- Changes in one workspace appear in all affected workspaces, KPIs and exception views.
- Local data survives hard reload; prototype reset is explicit and repeatable.
- All simulated AI/connectors are labelled; no false claim of live integrations or production security.
- Role switching demonstrates role-specific experience but must not be represented as server-enforced permissions.
- No broken links, undefined labels, placeholder values, contradictory dates or unsupported quantities.

17. Release roadmap

17.1 Roadmap view

Stage	Scope	Exit condition
Stage 0 — Full-coverage prototype	All 12 workspaces, 26 screens, 89 entities represented, approximately 920-field coverage mapping, 12 demo roles (corrected from 10), realistic linked seed data, simulated AI and local persistence.	A founder and representative employees can navigate the full product and complete simulated end-to-end activities. No production backend/security claims.
Pilot Wave 1 — First daily value	Assisted setup; company/sites/users/roles; supplier directory; machine register; central document vault; open jobs and configurable stages; basic store receipts/issues; breakdown logging.	Pilot users can perform daily updates; owner can see current pilot data and audit trail.
Pilot Wave 2 — Traceability and control	PR, supplier quote/PO/commitment, GRN, lot/certificate traceability, allocations, quality checks, rework, preventive maintenance and dispatch.	Material and document traceability demonstrated across representative jobs; quality and dispatch controls work.
v1.1+ — Expand after adoption	Richer RFQ/email intake, advanced remnants, stock reorder automation, richer maintenance, additional reporting and selected integrations.	Promote only after usage, data quality, customer value and willingness to pay are validated.
Strategic platform expansion	Collections/finance, people/expenses, unified inbox, tenders, WhatsApp/email orchestration and cross-industry capability packs.	Separate product-owner prioritisation and discovery; not silently included in manufacturing MVP.

17.2 Build sequencing note

Entity definitions, screen shells and workflows need not be implemented in the same order. Create screens and lightweight models early when useful, but only wire cross-entity workflows when dependent entities and service operations exist. The earlier S0–S9 implementation estimate of approximately 49 working days is a planning reference, not a committed delivery date or a reason to narrow the full prototype objective.

18. Acceptance criteria and test scenarios

ID / area	Pass condition
AC-01 Sales → Job	Create RFQ, versioned quote and customer PO; convert PO lines to one or more jobs. Verify source lineage, drawing revision, quote snapshot and remaining quantity after partial fulfilment.
AC-02 Procurement	Create PR linked to job/material need; capture supplier quotes; compare offers; submit supplier PO with required approval; revise commitment and surface impacted jobs.
AC-03 GRN and inventory	Receive a supplier PO partially; record received/accepted/rejected/hold separately; create lot(s); verify only accepted stock is available and PO outstanding balance remains correct.
AC-04 Stock ledger	Allocate stock to a job; verify available quantity reduces. Issue/return/transfer/adjust and confirm ledger-derived quantities; test reversal and transfer receipt.
AC-05 Dual UOM	Record primary and secondary quantities for representative plate, pipe/bar and sheet items using item-specific conversion parameters; verify no universal conversion is applied.
AC-06 Traceability	From a job, trace allocation to lot, supplier, PO, GRN and certificate. From a certificate, navigate to linked lots/jobs subject to permissions.
AC-07 Production stage	Record stage update with good/rejected/rework/scrap quantities; validate balance; display performer employee separately from logged-by user.
AC-08 Quality / NCR	Run incoming/in-process/final inspection; fail a check; raise linked NCR; create rework/reinspection; ensure unresolved quality state blocks release/dispatch.
AC-09 Asset breakdown	Report breakdown from mobile; link affected jobs; create maintenance event; consume spare through stock movement; record return-to-service and downtime.
AC-10 Dispatch	Create multi-job dispatch; select lines/lots; check QualityRelease and document pack; confirm dispatch; verify finished-goods stock-out and POD/delivery record.
AC-11 Document Vault	Upload, classify, version and link a document; set expiry obligation; verify reminder and role-based download logging.
AC-12 Business Pulse	Create or resolve shortage, overdue supplier, machine, QC and document situations; verify correct prioritisation, source links and updated status/counts.
AC-13 Reports	Verify reports and KPIs reconcile to source jobs, stock, supplier commitments, quality, maintenance and dispatch. No stale/inconsistent status enum mappings.
AC-14 Approval and RBAC	Verify requester cannot self-approve where maker-checker applies; rejection requires reason; Operator cannot approve/export; Auditor is scoped read-only.
AC-15 Persistence	Create dispatch, raise/close NCR, update stage, change status and resolve exception; hard refresh; verify every change remains and linked views still reconcile.
AC-16 Responsive	Complete GRN, stage update, inspection, breakdown and POD flows at 390px; complete operational review at 768/1024 and 1440px.
AC-17 Prototype honesty	Verify simulated AI, integrations, security and sync are labelled accurately. No UI message claims a live provider execution without a confirmed result.

19. Open decisions, assumptions and risks

Open item	Why it matters / required validation	Owner
Exact manufacturing process and product types	Walk through recent real jobs, product families, routing and top operational pain.	Pilot owner / production head
Material/UOM model	Validate stock registers and actual weight/length/area/sheet conversion rules for each item family.	Stores / engineering
MTC/COA semantics	Confirm certificate is per heat, lot, shipment or item; verify how quality maps it to material receipt.	QC / procurement
Non-PO jobs	Confirm whether verbal, LOI, internal, repair or make-to-stock jobs are needed in the pilot and what authorisation applies.	Owner / Sales
Customer PO delivery schedules and QAP	Confirm multi-shipment schedules, customer inspection clauses and third-party witnessing requirements.	Sales / QC
Outward processing / subcontracting	Confirm whether material leaves the site for cutting, plating, heat treatment or other job work.	Production / procurement
Customer-owned material	Confirm whether customer-furnished material is received and must be ownership-scoped.	Stores / owner
Shared-device attribution	Decide whether operators have own logins, PIN/badge attribution or supervisor-recorded employee identity.	Production / IT
Maintenance meter model	Confirm calendar-only, run-hour, cycle or mixed schedules and how readings are captured.	Maintenance
Tally/accounting system of record	List existing tools and decide which system owns customers, invoices, receipts and payment status.	Owner / accounts
Offline behaviour	Confirm devices/connectivity and acceptable draft-only versus queued write behaviour; no consequential auto-execution on reconnect.	Pilot team / engineering
AI provider, budget and data processing	Approve whether any live extraction call is enabled, which provider, data retention and monthly spend cap.	Product / engineering
Languages and shop-floor literacy	Validate required labels, voice input and local language support with actual users.	Pilot team
Commercial model and willingness to pay	Test value proposition, onboarding effort, support cost and pricing with design partners.	Product / business
Field Dictionary v0.2	Attach and reconcile the actual v0.2 workbook against the 89-entity v1.1 scope before schema freeze.	Product owner / engineering

19.1 Key risks

- Overbuilding before validating customer workflow could increase complexity without proving adoption or willingness to pay.
- Incorrect UOM conversions or unreliable opening balances can undermine trust in inventory data.
- Weak data capture on the shop floor may cause stale job progress and false operational insights.
- AI-generated insight without clear source/provenance can create unsafe confidence and erode user trust.
- Prototype role switching or local storage could be mistaken for production security or resilience unless clearly labelled.
- Combining the broad cross-industry BRD directly into the manufacturing pilot would expand scope significantly; separate roadmap governance is required.

Appendix A — Screen and workspace completeness checklist

Use this checklist during prototype acceptance. “Implemented” means the screen renders, its primary controls work, its data relationships resolve and its main create/update path persists locally. A route shell alone is not complete.

ID	Screen	Acceptance checklist
UX-01	Home / Business Pulse	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-02	RFQ & Quotation	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-02b	Customer PO & Job Creation	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-amend	Customer PO Amendment	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-03	Supplier Workspace	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-04	PR / Supplier RFQ / Quote Comparison	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-05	Supplier PO & Receipt	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-05b	Supplier Return	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-05s	GRN / Goods Receipt	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-06	Inventory Overview	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-07	Material Lot Detail	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-cert	Certificate Extraction	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-transfer	Stock Transfer	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-cycle	Cycle Count	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-remnants	Remnants & Scrap	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-08	Job Board	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-09	Job Workspace	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-10	Stage Update	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-11	Quality Inspection	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-ncr	NCR / Rework Workspace	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-12	Assets & Maintenance	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-dispatch	Dispatch	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-13	Document Vault	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-reports	Reports & Insights	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-14	Admin & Security	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access
UX-15	Audit & Activity Timeline	[ ] View/form [ ] Linked data [ ] Core action [ ] Responsive [ ] Role access

Appendix B — Entity catalogue by dependency layer

The Scope Matrix v1.1 lists 89 entities grouped into eight dependency layers. This catalogue is a coverage reference; the companion field dictionary remains the source for exact field definitions, types, requiredness, validation and relationships.

Layer 0 — Platform / foundation (15)

Entity		
Tenant	Company	LegalEntity
Site	StoreLocation	User
Membership	Role	RoleGrant
ApprovalPolicy	ApprovalRequest	AuditEvent
Notification	UOM	MaterialCategory

Layer 1 — Master data (18)

Entity		
Party	PartyContact	PartyAddress
PartyGSTRegistration	MaterialItem	MaterialAttributeDefinition
SupplierMaterial	Asset	WorkCentre
Document	DocumentVersion	DocumentLink
DocumentObligation	StageTemplate	StageTemplateStep
InspectionTemplate	InspectionTemplateCheck	Employee

Layer 2 — Commercial transactions (16)

Entity		
CustomerRFQ	RFQLine	CustomerQuotation
CustomerQuotationLine	CustomerPO	CustomerPOLine
CustomerPOAmendment	SupplierRFQ	SupplierRFQLine
SupplierQuotation	SupplierQuotationLine	SupplierPO
SupplierPOLine	SupplierPOAmendment	SupplierCommitment
PurchaseRequisition		

Layer 3 — Job and production (7)

Entity		
Job	JobLine	JobMaterialRequirement
PurchaseRequisitionLine	JobStage	JobUpdate
JobBlocker		

Layer 4 — Inventory transactions (9)

Entity		
GoodsReceipt	GoodsReceiptLine	MaterialLot
StockMovement	MaterialAllocation	StockTransfer
StockTransferLine	StockCount	StockCountLine

Layer 5 — Quality and dispatch (10)

Entity		
MaterialCertificate	CertificateLotLink	Inspection
InspectionResult	NCR	ReworkOrder
QualityRelease	Dispatch	DispatchLine
DispatchLotLink		

Layer 6 — Maintenance and dispatch completion (8)

Entity		
MaintenancePlan	MaintenanceOccurrence	MaintenanceEvent
MaintenanceEventJobLink	MaintenanceSpareConsumption	AssetMeterReading
DeliveryConfirmation	InvoiceStatus	

Layer 7 — AI, import and supplementary (6)

Entity		
AgentRun	AgentActionProposal	ImportJob
SupplierReturn	PaymentStatusEvent	DispatchDocumentPack

Appendix C — Prototype seed-data baseline

Seed group	Target baseline / sample contents
Sites	Hyderabad Factory (site-hyd); Hyderabad Warehouse (site-wh).
Store locations (8)	Raw Material Store (RM); WIP Store (WIP); Finished Goods Store (FG); Quarantine Store (QRN); Spare Parts Store (SPR); Dispatch Bay (DSP); Warehouse Bay A; Warehouse Bay B.
Users (10)	Ravi Shankar — Owner/GM; Anita Mehta — Sales; Karthik Raju — Buyer; Asha Nair — Storekeeper; Suresh Kumar — Production Supervisor; Mohan Das — Operator; Priya Patel — QC; Venkat Rao — Maintenance; Deepa Singh — Dispatch; External Auditor — Auditor.
Employees (10)	Mohan Das, Suresh Kumar, Venkat Rao and seven shop-floor/helper employees. Some employee records have no linked EOSS User account.
Customers (6)	Bharat Power Equipment; NTPC Vendor Division; Meridian Engineering; L&T Hydrocarbon; Anand Fabricators; Sri Venkateswara Industries.
Suppliers (8)	MSK Steel; Stainless Source India; RK Fasteners & Hardware; Bharat Gas Supply; Apex Consumables; SteelCraft Tubes; Precision Components; Metro Scrap Dealers.
Materials (15)	MS Plate 8mm, MS Plate 12mm, MS Pipe 100NB Sch40, SS Sheet 3mm 304L, MS Flat Bar 50x6, welding rods, MIG wire, bolts, nuts, Argon, CO2, grinding discs, primer paint, anti-spatter spray and electrode holder spare.
Assets (8)	Two operational welding machines, one welding machine in maintenance, plasma cutter, press brake, overhead crane, angle grinder and drill press in breakdown.
Jobs (8 total)	J-101 through J-108. Use seven active/held/working jobs and one completed/dispatched reference job; do not label all eight as active.
Material lots / stock	Maintain valid lot records only for lot-tracked items. Welding rods with lot tracking disabled remain aggregate stock. The SS Sheet ordered but not received is a supplier commitment, not a MaterialLot.
Supplier POs (5)	Mix of on-time, overdue, partially received and fully received supplier POs; include commitments and an amendment on the overdue SS Sheet PO.
Documents (10)	ISO 9001 certificate, supplier quality certificate, two MTCs, customer drawings, factory licence, inspection report, gas COA and machine calibration certificate.
Pulse situations (8)	3 critical, 3 warning and 2 informational examples across material, procurement, asset, quality and document categories.

C.1 Seed-data corrections and consistency rules

- Count jobs as eight total with one completed/dispatched reference job, not eight active jobs.
- LOT-010 (SS Sheet on order) must be represented as an expected supplier receipt/commitment until the GRN creates an actual lot.
- LOT-007 (welding rods with lot tracking disabled) should be represented as aggregate stock, not an artificial lot.
- Use a single deterministic DEMO_NOW reference across relative timestamps, overdue labels, Pulse, reports and workspaces.
- Ensure the inventory screen surfaces all eight locations. A screenshot showing four locations is a UI/data presentation gap, not a scope reduction.
- Ensure all KPI counts, PO/job/lot/asset references and approval labels resolve to actual seeded records; never show “undefined”.
- For dual-UOM stock, reconcile primary and secondary values using the item-specific conversion method/parameters.

Appendix D — Reusable component inventory

Group	Components
Shell & navigation	AppShell; Sidebar; BottomNav; TopBar; SubNav; RoleSwitcher.
Data display	DataTable; FilterBar; RecordHeader; StatusChip; DetailDrawer; MovementLedger; StageTimeline; LotTraceCard; DocumentCard; ExceptionCard; ActivityFeed; MobileCard.
Forms & input	SchemaForm; QtyInput; EntityPicker; StageQtyInput; InspectionChecklist; CertificateUploader; FileUpload.
AI & approvals	AIProposalBanner; AISimBanner; ApprovalRequestCard; AskEOSSPanel.
Feedback & navigation	ConfirmModal; WizardShell; Toast; InContextAlert; EmptyState; SkeletonRow; NotificationDropdown; OfflineDraftBadge.

Total: 37 components as stated in Scope Matrix v1.1. Component contracts should remain consistent across workspaces, but each workspace may compose them differently to suit the task.

Appendix E — Source register and governance notes

Source	Date / version	Use in this PRD
EOSS Pilot Product Definition v0.1	3 Nov 2026	Manufacturing pilot scope, product promise, workflows, capability depth, data rules, roles, security and acceptance criteria. Working baseline.
EOSS Pilot Product Architecture & UX Blueprint	3 Nov 2026	53-page manufacturing UX/architecture blueprint: 51 review-gap reconciliation, experience principles, IA, workflows, role matrix, screens, design system and AI patterns.
EOSS Pilot Definition Review	3 Nov 2026	Independent review identifying 51 schema/operational gaps. Accepted corrections should be reflected in v1.1 scope.
EOSS Implementation Clarification Addendum	3 Nov 2026	Entity dependency graph, role/permission clarification and build-sequencing correction. Original four-week demo milestone is not the current full-prototype scope.
EOSS Scope Matrix v1.1	4 Nov 2026 (current shared version)	Current full-coverage manufacturing prototype baseline: 12 workspaces, 26 screens, 89 entities, approximately 920 fields, 37 components, 10 roles, seed data and build sequence.
EOSS Field Dictionary v0.1	Earlier working version	Older workbook with detailed entities/fields. Superseded where changed by accepted review corrections and v0.2.
Field Dictionary v0.2	Referenced by Scope Matrix v1.1; not present in assembled file set	Required for final field-level traceability and production schema freeze. Attach/reconcile before final data model approval.
Enterprise OS BRD v2.0	1 Nov 2026	Broader cross-industry Business OS vision and roadmap, including inbox, tenders, finance/collections, people and industry packs. Strategic reference; not manufacturing MVP scope by default.
Enterprise OS Product Experience Model & Detailed UX Blueprint v0.3	3 Nov 2026	Broader solo-founder/owner-operator UX model with Pulse/Companion, progressive complexity, multilingual/mobile/offline and cross-industry flows. Reuse compatible principles; manufacturing pilot decisions take precedence.
EOSS Product Discovery: Indian SME Manufacturing	3 Nov 2026	Discovery research and initial manufacturing workflow hypotheses. Meeting-derived and industry-inference items require pilot validation.
Factory discussion / product-owner conversations	3–4 Nov 2026	Current product mantra, full-coverage prototype objective, UI feedback and visual direction. Product-owner decisions govern over older defaults where conflicting.

E.1 Governance decisions

- This PRD is a consolidated product requirement baseline, not a replacement for the detailed field dictionary, screen-level interaction specification or production architecture decision records.
- When the field dictionary v0.2 becomes available, reconcile all 89 entities, relationships, field definitions, validation rules and screen mappings. Record differences rather than silently changing them.
- Any expansion into cross-industry HR/payroll, finance/collections, tender, unified inbox or advanced integrations requires explicit scope decision, user validation and a separate release priority.
- Any live AI provider, messaging connector, accounting connector or production security claim requires explicit technical, security, privacy and cost approval.

PRD sign-off and next decisions

This PRD is ready for product-owner review as a consolidated baseline. It deliberately separates the broader EOSS platform ambition from the manufacturing-first MVP and the full-coverage prototype that is currently being built.

Decision	Status
Product mantra and user promise	Agreed direction: easy data entry, effortless data collection, intelligent automation, streamlined daily operations.
Manufacturing-first initial vertical	Current product focus; exact factory process and customer validation remain open.
Full prototype coverage	Scope Matrix v1.1 baseline: 12 workspaces, 26 screens, 89 entities, approximately 920 fields and 10 roles.
Visual direction	Dark-first Industrial Intelligence reference direction requested; final design system not yet approved.
Field Dictionary v0.2	Referenced by Scope Matrix v1.1 but requires attachment/reconciliation before production schema freeze.
Production pilot scope and priorities	To be validated with the design-partner manufacturer after the full prototype is experienced.
Security, privacy and live integrations	Production gate; no compliance or live connector claim is made by the prototype.

DEFINITION OF SUCCESS

A solo founder can understand the business situation quickly, employees can complete common tasks with minimal data entry, operational information is captured once and reused across the business, and EOSS intelligently surfaces the next action without taking consequential decisions away from people.

20. Binding product-owner clarifications — v1.1

This addendum resolves the twelve review comments received after PRD v1.0. It is part of the PRD and takes precedence over conflicting v1.0 wording, counts, assumptions or screen descriptions. The original manufacturing-first scope is preserved; clarifications make organisation structure, stock fulfilment, traceability, procurement evidence and screen mapping explicit.

IMPLEMENTATION RULE

Claude or any implementation agent must read the complete base PRD and this addendum together. Implement all 12 workspaces and all 27 screen identifiers using the approved Aurora Indigo Home design language. Do not stop after the Home screen, a prototype shell, or a planning report.

20.1 Revised baseline and precedence

Dimension	v1.1 binding baseline
Primary vertical	Indian SME manufacturing, industrial fabrication and trading/distribution workflows where relevant.
Workspaces	12; existing navigation structure is retained.
Top-level screen identifiers	27. One screen, UX-02a Customer Directory, is added.
Planned domain entities	94. The original 89 remain the baseline; five entities are added or formalised below.
Reusable components	37 planned patterns remain the reusable component baseline; do not increase the component count just to create new names for existing patterns.
Roles	12 demo roles (corrected from 10) as defined in the base PRD.
Field coverage	Approximately 920 as a planning estimate only. Reconcile against the actual Field Dictionary v0.2 before production schema freeze.
Visual baseline	Aurora Indigo is the approved default Home theme. Use the approved current Home screenshot as the visual source of truth across all workspaces.

Where the base PRD states 26 screens or 89 entities, the v1.1 values above supersede those counts. Existing screen IDs remain unchanged except for the added UX-02a. The full screen-to-module map appears in section 20.10.

20.2 Organisation hierarchy: tenant, business group, company, GST registration, site and store

EOSS must support a single business with multiple operating sites as well as a parent business group containing multiple companies. These are different levels and must not be collapsed into one generic “location” or “company” field.

Level / record	Meaning in EOSS	Relationship and behaviour
Tenant / Workspace	The SaaS customer account and hard data-isolation boundary.	Owns the EOSS workspace, subscription and tenant-level settings. A user must never see another tenant’s data.
BusinessGroup (new)	Optional parent-group container for related companies or operating businesses.	One tenant may have one or more groups. A group may contain multiple Companies. It provides authorised group-level overview and

Level / record	Meaning in EOSS	Relationship and behaviour
		administration; it does not merge legal ownership or stock balances.
Company	The operating business profile users select when working in EOSS.	Belongs to a BusinessGroup when group structure is used. References one LegalEntity for the normal pilot model. A group with three separately incorporated companies has three Company records.
LegalEntity	The underlying registered legal person and PAN-level identity.	Stores legal name, PAN, entity type and statutory identity. One legal entity may have multiple GST registrations.
CompanyGSTRegistration (new)	One GSTIN registration record for a LegalEntity.	Store GSTIN, state code, registration type/status, effective dates, principal place and related evidence. Do not assume one PAN always equals one GSTIN.
Site	A physical operating premises such as a factory, branch, office or warehouse.	Belongs to one Company. A Company can have multiple Sites. A Site is not an internal stock bin and is not automatically a separate company.
SiteGSTRegistration (new)	Link between a physical Site and a GST registration.	Supports the primary/default GST registration and additional valid GSTIN associations where needed. Preserve effective dates and the principal/additional-place context.
StoreLocation	An internal stock-holding area within a Site, such as Raw Material, WIP, Finished Goods, Quarantine, Spare Parts or Dispatch Bay.	Belongs to one Site. Multiple StoreLocations can exist inside one Site. It controls stock placement and movements, not legal identity.

SIMPLE EXAMPLE

One manufacturing company with a Hyderabad factory and Hyderabad warehouse = one Company, one LegalEntity, one or more relevant GST registrations, two Sites, and multiple StoreLocations under each Site. A parent group with three legally separate companies = one BusinessGroup with three Company records, each with its own LegalEntity and applicable GST registrations.

GST terminology matters: a GSTIN is a tax registration, not proof that there is a separately incorporated company. Under CGST Act section 25(4), each registration obtained or required by the same person is treated as a distinct person for GST purposes. The model must therefore allow one LegalEntity/PAN to have multiple GSTIN records, while separately incorporated entities remain separate Company/LegalEntity records.

- A Company can have multiple Sites; a Site can have one default GSTIN and additional GSTIN links where the actual registration facts require them.
- The site selector filters operational records. The company/group selector changes the legal/operating business context. The StoreLocation selector filters stock within the selected Site.
- A group-level dashboard may show consolidated company summaries only for users granted group scope. It must retain company, GSTIN and site drill-down and must not present group stock as freely interchangeable.
- Same-company transfers between Sites use StockTransfer with source/destination company, GSTIN, site and store references. Cross-GSTIN or cross-company transfers must be flagged as inter-registration/inter-company movement, capture required document references, and follow configured approval. They must not silently merge stock ledgers.

- The manufacturing MVP records transfer evidence and references (for example invoice/challan/transport document reference where relevant); it does not calculate GST, generate statutory invoices or generate e-way bills unless separately approved.

Recommended hierarchy: Tenant → BusinessGroup (optional) → Company → LegalEntity → CompanyGSTRegistration; Company → Site → StoreLocation; Site ↔ GST registration through SiteGSTRegistration. Users and grants can be scoped to tenant, group, company, GST registration, site and store as applicable.

20.3 Customer master is a separate first-class directory

The customer directory must have its own main table and customer detail experience. Do not bury customer master data only inside RFQs or reuse the supplier list as the customer screen.

- Retain Party as the canonical shared legal/trading-party identity and common contacts/addresses. Add CustomerAccount as a customer-specific profile linked to Party; the same Party may hold both CustomerAccount and supplier classification if the business relationship genuinely works both ways.
- CustomerAccount holds customer code, segment/type, status, assigned salesperson, payment terms, credit/limit settings where enabled, default bill-to and ship-to preferences, GST registration links, notes and activity summary.
- UX-02a Customer Directory provides a searchable/filterable customer table, customer detail, contacts, addresses, GST details, RFQ/quotation/PO history, open jobs, dispatch/delivery history, documents and activity.
- Customer identity is captured once and re-used by RFQ, quotation, customer PO, job, dispatch and delivery flows. Editing shared legal identity must be permission-controlled and audited.

New entity: CustomerAccount. Existing Party, PartyContact, PartyAddress and PartyGSTRegistration remain in use for shared party identity and external party registration data.

20.4 Finished goods belong in Inventory, not Assets

Completed products intended for sale or customer delivery are inventory. Store them in the Inventory & Stores workspace, normally in a Finished Goods StoreLocation, with item/specification, quantity/UOM, lot/batch (where configured), QC/release state, location, reservation and movement history.

Assets is for durable equipment/resources the business uses to operate—machines, cutters, cranes, vehicles, tools or other maintainable equipment. Assets have an asset register, work-centre/site assignment, maintenance plan, downtime and service history. A manufactured item does not become an Asset simply because production is complete. If a business later capitalises an item for its own use, that is a separate accounting/fixed-asset process outside this manufacturing MVP.

20.5 Customer PO evidence: upload the document received from the customer

Extend FR-SALES and UX-02b to accept a customer-issued Purchase Order as original evidence, not just manually typed PO fields. In normal sales terminology, a Customer PO is issued by the customer to EOSS's customer-company; a supplier normally sends a Supplier Quotation or acknowledgement. The UI must label these document types clearly.

- Allow PDF upload, image/screenshot attachment and mobile camera/photo capture; support multiple pages/files and preserve the original file.
- Capture received date, source/channel, uploader, document number if known, version/revision and the CustomerPO link. Keep original evidence immutable; new revisions create new document versions.
- AI/OCR may suggest customer, PO number, dates, lines, quantities, rates, terms and drawing references, but extracted values remain a proposed draft until a user reviews and confirms them.
- Compare the confirmed PO with the accepted quotation and display differences in quantity, price, specification, drawing revision, delivery schedule and terms; route material changes for review as required.

If the uploaded file is a supplier-originated quotation, attach it to SupplierQuotation in Procurement instead of misclassifying it as a CustomerPO.

20.6 Supplier quotations must be captured as first-class procurement records

Strengthen FR-PROC and UX-04: every supplier quote received in response to a Supplier RFQ must be stored as an individual SupplierQuotation record and must remain traceable even if it is not selected.

- Capture supplier, source SupplierRFQ and line(s), received date, quote/reference number, currency, validity/expiry, quoted quantities, UOM, rate, taxes, freight, discount, MOQ, lead time, delivery promise, payment terms and notes as available.
- Attach the original supplier quotation PDF, spreadsheet, email file or image. Preserve versions and source provenance.

- Compare supplier offers side-by-side on a normalised like-for-like basis. Explicitly mark values that are missing, expired or not comparable; never silently assume zero freight/tax or convert incompatible units.
- Record selected, rejected, shortlisted or expired status and an optional decision reason. A SupplierQuotation is an offer, not a SupplierPO and not inventory.

The existing SupplierQuotation and SupplierQuotationLine entities remain authoritative; the enhancement is full evidence capture, workflow, comparison and traceability, not a duplicate quote entity.

20.7 Job material requirements: show exactly what is needed to complete each job

Expand the Job Workspace Materials area into a practical job material plan / material take-off. The screen must answer: what material does this job require, what is already available, what is reserved or issued, what is still short, and what needs to be purchased or received before each stage can run?

- For each JobLine and relevant JobStage, record MaterialItem/specification/grade/dimensions, required quantity and UOM, planned quantity, required-by date/stage, and source of requirement (template/BOM, drawing take-off, manual authorised entry or approved revision).
- Show on-hand, reserved, available, QC-held/blocked and in-transit amounts separately by Site/StoreLocation and lot where applicable. Available is calculated by rules; it is never manually typed.
- Track reserved/allocated, issued, returned and consumed quantities through MaterialAllocation and StockMovement references. Show remaining-to-issue and shortage quantities as derived values; do not create duplicate free-text stock balances.
- Allow a shortage to create a linked PurchaseRequisition with required date, site/store, priority, quantity/UOM and affected job/stage. Link the resulting supplier RFQ, SupplierQuotation, SupplierPO, GRN and MaterialLot back to the requirement.
- Allow approved substitutions only after material/specification compatibility checks and human approval. Preserve the original requirement and substitution reason.

The Job Workspace must surface material readiness at job level and at stage level. A stage that requires unavailable/held material should show the blocker and reason; a permitted supervisor may record an authorised exception with an audit trail.

20.8 Ready-stock fulfilment: allow delivery without unnecessary production

The Order-to-Delivery process must be stock-first where appropriate. Not every customer order requires a new Job or a production stage. The same Business OS must support make-to-order manufacturers, make-to-stock manufacturers and traders/distributors.

Scenario	Required behaviour
Ready stock fully covers order	Check sellable/available Finished Goods or trading stock; reserve against the CustomerPOLine; create/prepare Dispatch directly. Do not force creation of a production Job.
Ready stock partially covers order	Reserve the available quantity. Route the balance to a production Job for make-to-order manufacturing or a PurchaseRequisition/SupplierPO for buy-to-order trading. Keep the outstanding customer quantity visible.
No ready stock — manufacturer	Create or plan a Job, derive JobMaterialRequirements, identify procurement shortages and track stage execution through QC release and dispatch.
No ready stock — trader/distributor	Create procurement demand and buy/receive the goods; do not show production stages that are not relevant to the company/item configuration.
Partial dispatch/delivery	Dispatch only reserved/available quantities; retain the unfulfilled balance and subsequent fulfilment sources against the same CustomerPO line.

- Add SalesOrderStockAllocation to reserve ready stock for a CustomerPOLine by item, Site, StoreLocation and optional MaterialLot; capture quantity/UOM, allocation status and release/consumption references.

- DispatchLine must link to the relevant CustomerPOLine and fulfilment source. For stock fulfilment, link lot/stock allocations; for production fulfilment, link Job/JobLine and finished-goods movements.
- Do not treat in-transit supplier commitments, QC-held stock, blocked stock or unposted receipts as immediately deliverable ready stock.
- The Home Order-to-Delivery flow is context-aware: production may appear for manufacturers, while trading flows can move from Sales Orders through Stock/Procurement to Dispatch without a fake Production stage.

20.9 Purchase-order approvals are conditional, not universal

Approval is policy-driven. Do not require every SupplierPO to go through an approval queue. ApprovalPolicy determines whether a request needs approval based on configurable company/site/category/vendor/value/risk rules.

- Example rules may include value threshold, new/unapproved supplier, non-budgeted purchase, price variance above tolerance, restricted material, advance payment, unusual terms, negative-stock exception or cross-company/cross-GSTIN transfer.
- If a SupplierPO matches no approval rule and the creator has the required create/issue permission, it can proceed directly through the authorised issue path. Do not manufacture a dummy approval record.
- If approval is required, show pending/approved/rejected status and enforce maker-checker where configured. Rejection requires a reason; approval stores requester, approver, decision, timestamp, reason and action snapshot.
- All paths must respect permissions and be audited. A PO remains a supplier commitment only; it does not increase stock. Only accepted GRN quantities become usable stock.

20.10 Explicit module-to-screen mapping

Every screen must identify both its stable UX screen ID and its owning module/workspace ID. UX IDs are identifiers for acceptance tests and deep links; module IDs identify the navigation/permission/reporting domain. A screen may link to another module but must have one primary owner.

Module ID	Owning module/workspace	Screen ID	Screen name
MOD-01	Home / Business Pulse	UX-01	Home / Business Pulse
MOD-02	Sales & Customer Orders	UX-02	RFQ & Quotation
MOD-02	Sales & Customer Orders	UX-02a	Customer Directory (new)
MOD-02	Sales & Customer Orders	UX-02b	Customer PO & Job Creation
MOD-02	Sales & Customer Orders	UX-amend	Customer PO Amendment
MOD-03	Suppliers	UX-03	Supplier Workspace
MOD-04	Procurement	UX-04	Purchase Requisition / Supplier RFQ / Quote Comparison
MOD-04	Procurement	UX-05	Supplier PO & Receipt
MOD-04	Procurement	UX-05b	Supplier Return
MOD-04	Procurement	UX-05s	GRN / Goods Receipt
MOD-05	Inventory & Stores	UX-06	Inventory Overview
MOD-05	Inventory & Stores	UX-07	Material Lot Detail
MOD-05	Inventory & Stores	UX-cert	Certificate Extraction
MOD-05	Inventory & Stores	UX-transfer	Stock Transfer
MOD-05	Inventory & Stores	UX-cycle	Cycle Count
MOD-05	Inventory & Stores	UX-remnants	Remnants & Scrap
MOD-06	Jobs & Production	UX-08	Job Board
MOD-06	Jobs & Production	UX-09	Job Workspace

Module ID	Owning module/workspace	Screen ID	Screen name
MOD-06	Jobs & Production	UX-10	Stage Update
MOD-07	Quality	UX-11	Quality Inspection
MOD-07	Quality	UX-ncr	NCR / Rework Workspace
MOD-08	Assets & Maintenance	UX-12	Assets & Maintenance
MOD-09	Dispatch & Delivery	UX-dispatch	Dispatch
MOD-10	Document Vault	UX-13	Document Vault
MOD-11	Reports & Insights	UX-reports	Reports & Insights
MOD-12	Admin & Security	UX-14	Admin & Security
MOD-12	Admin & Security	UX-15	Audit & Activity Timeline

The Customer Directory is a new screen within the existing Sales & Customer Orders workspace; it does not create a thirteenth workspace. The resulting baseline is 12 workspaces and 27 top-level screen identifiers.

20.11 Change history and audit trail must be persisted

The application must be able to answer “who changed what, when, from which record/context, and what was the previous value?” This is a data integrity requirement, not merely a visual activity feed.

- Mutable domain records should carry created_at, created_by_user_id, updated_at and updated_by_user_id where applicable. Use immutable IDs and timezone-aware timestamps.
- AuditEvent remains the append-only canonical audit record. Capture actor/user, actual performer employee where relevant, tenant/company/GSTIN/site scope, action, entity type/record ID, timestamp, source (UI/import/API/AI/system), before/after field diff, reason, correlation/request ID and approval reference when applicable.
- Audit consequential create/update/status/approval/rejection, record revision, posted quantity/stock movement, allocation/release, QC hold/release/NCR, dispatch/POD, role/permission changes, document upload/version/download and inter-company/inter-GSTIN transfer events.
- Do not log secrets, authentication tokens or unnecessary personal data. Restrict audit visibility by permission; auditors receive scoped read-only access.
- Production audit writes must be persisted server-side in the same transaction as the domain change where possible, using service/domain event emission and database-level safeguards/triggers as appropriate. UI-only logs are insufficient. Prototype audit events must persist locally and survive hard refresh.
- Posted stock movements, issued quote versions and approved action snapshots are append-only or versioned; corrections use reversal/amendment records rather than silent overwrite.

20.12 Approved Home visual system and AI/search behaviour

The latest user-approved Antigravity Home screenshot is the visual source of truth. Use the Aurora Indigo theme across all workspaces: deep indigo navigation, lavender page surface, white cards, restrained purple primary actions, green success, amber warning, red error, clear typography and soft shadows. Keep existing theme token architecture where implemented, but Aurora Indigo is the default and the only theme to use for this implementation unless the product owner explicitly requests a theme comparison.

Element	Binding UX rule
Global search (top bar)	Record discovery and navigation only. Search customers, suppliers, RFQs, quotations, POs, jobs, materials, lots, documents and employees. Support Ctrl+K/Cmd+K, recent searches, keyboard navigation and permission-aware results. It is not a conversational assistant.
EOSS Copilot (right panel)	The one conversational assistant. It explains data, answers

Element	Binding UX rule
	operational questions, identifies risks, suggests actions, prepares drafts and presents proposals for confirmation. It is not a second search box.
Copilot robot visual	Remove the patchy/rectangular colour block or mismatched halo visible behind the robot asset. Prefer a transparent asset or a clean mask; place the robot on the same continuous panel gradient with only a subtle, smooth radial glow that matches the panel. No hard-edged image background, visible image rectangle, patchy texture or clashing backdrop.
Home top KPI row	One row of financial/stock KPIs only: Sales, Purchases, Stock Value, Receivables, Payables and Profit. Each underlying measure appears once in the KPI row.
Order to Delivery	Operational stage counts only: Sales Orders, Material shortages/ready stock, Production in progress (when relevant), Quality pending (when applicable) and Dispatch ready. Do not repeat Sales/Purchases/Stock Value/Profit or repeat the same production/dispatch total KPI elsewhere.
Lower Home panels	Priorities, order/job-level production progress, Business Health factors and Cash Flow. Use source-linked details; avoid duplicating top KPI amounts or stage counts.
Bottom whitespace	Do not stretch cards or enlarge rows unnaturally to fill empty viewport space. Add a compact Quick Actions strip below the lower Home panels with role-aware actions such as Create Sales Order, Create Purchase Request, Receive Stock, Update Job, Record Inspection and Plan Dispatch. Actions must be functional and hidden/disabled according to permissions. At narrower heights allow normal scroll; preserve useful spacing.
Copilot position	On desktop, use a clearly visible, full-height right-side panel spanning the dashboard content area. On mobile/tablet, provide a collapsible panel or accessible drawer/bottom sheet; do not squeeze the content or obscure primary tasks.
Business type adaptation	Use the same design system for manufacturers and traders, but configure the operational flow. Manufacturers may show Production and Quality; traders may show stock/procurement/dispatch without fake production steps.

20.13 Full-application design/build instruction

The implementation is not limited to the Home screen. All 27 screen identifiers and all 12 workspaces must be built using this same approved design language. Do not generate 27 copies of the Home dashboard. Each workspace must use the layout best suited to its task: dense but readable tables for registers; record detail with tabs and relationship panels; progressive forms/wizards for data capture; timelines for traceability; checklists for QC; mobile task flows for shop-floor work; and focused review/approval surfaces for decisions.

- Maintain consistent app shell, sidebar, top bar, typography, spacing, controls, status chips, drawers, dialogs, notifications, loading/empty/error states and Copilot entry point across all screens.
- Preserve the 37 reusable component patterns in Appendix D and compose them differently per workspace. Workspace-specific composition is required; repeating identical KPI-plus-table layouts is not acceptable.
- Implement all visible actions, filters, tabs, table rows, drawers, upload controls and state transitions. Do not leave decorative controls or toast-only state-changing actions.

- Use one shared source of truth for canonical records, relationships, quantities and statuses across all workspaces. Any mutation must update every affected view, relevant Home situation and report.
- For each role, render the appropriate task-first experience; do not expose approval/export actions that the role cannot perform. Client-side hiding in the prototype is a demonstration only, not production security.

20.14 New/updated entity baseline

Entity	Status	Purpose
BusinessGroup	New	Optional parent grouping of related Company records, with group-level access and reporting scope.
CompanyGSTRegistration	New	Company/LegalEntity GSTIN registration record with state, type, status and effective dates.
SiteGSTRegistration	New	Effective-dated link between a Site and the GST registration(s) valid for that premises.
CustomerAccount	New	Dedicated customer profile and directory record linked to canonical Party identity.
SalesOrderStockAllocation	New	Reservation of ready stock against a CustomerPOLine, with site/store/lot and quantity lineage.
StockTransfer	Existing; fields/behaviour clarified	Add company/GSTIN/site scope and intra-company vs cross-GSTIN/inter-company transfer handling.
AuditEvent	Existing; requirements strengthened	Append-only audit record with actor, action, scope, source and before/after diff.
JobMaterialRequirement	Existing; requirements strengthened	Job/stage-level material take-off and readiness, linked to allocations, stock movements and procurement.
SupplierQuotation / SupplierQuotationLine	Existing; requirements strengthened	Persist each received supplier offer, evidence, validity and comparison outcome.

Planned entity count: 94 (89 original baseline + five new entities). Confirm exact table/field definitions and the v0.2 field dictionary before production schema freeze. Do not implement many-to-many relationships as undocumented JSON ID arrays when a link entity carries business meaning or quantity/provenance.

20.15 New acceptance tests

ID	Pass condition
AC-18 Organisation hierarchy	Create a tenant with one BusinessGroup and multiple Companies; create multiple Sites/StoreLocations per Company; verify company/site/store navigation and group-level visibility by role.
AC-19 GST model	Create one LegalEntity/PAN with multiple GSTIN registrations and separately incorporated Companies with different legal identities. Verify GSTIN/site mapping and no false assumption that every GSTIN is a separately

ID	Pass condition
	incorporated company.
AC-20 Inter-company / GST movement	Transfer stock within a company and across GSTIN/company boundaries. Verify source dispatch, in-transit state, destination receipt, linked document references, approval and separate ledgers; no automatic tax calculation.
AC-21 Customer master	Create and edit CustomerAccount in UX-02a; verify customer directory is separate from Supplier Workspace and shared Party identity is reused in RFQ/quotation/PO/job.
AC-22 Customer PO evidence	Upload PDF and mobile photo/screenshot to CustomerPO; preserve original/version; verify extracted values require human confirmation and comparison against accepted quotation.
AC-23 Supplier quote traceability	Capture multiple SupplierQuotations with attachments, validity and line pricing; compare like-for-like; select/reject with reason; ensure SupplierPO remains separate.
AC-24 Job material plan	Create JobMaterialRequirements; verify ready/reserved/issued/returned/consumed/QC-held/in-transit quantities; create linked PR from shortages; trace through PO, GRN and lot.
AC-25 Ready-stock delivery	Fully fulfil a CustomerPO line from available finished/trading stock without creating a Job. Verify partial stock plus production/procurement balance and direct dispatch links.
AC-26 Conditional PO approval	Test a PO under a policy requiring approval and a PO that does not match any approval policy. Verify only the first enters approval; both remain permission-checked and audited.
AC-27 Audit history	Change a record and a consequential status/quantity; verify actor, timestamp, source and before/after diff persist after reload and are visible only to permitted roles.
AC-28 Finished goods vs assets	Complete production and confirm output enters Finished Goods Inventory; verify machines/equipment remain in Assets and maintenance.
AC-29 Screen/module traceability	Verify every one of the 27 UX IDs maps to exactly one primary MOD ID, route, permission domain and acceptance test.
AC-30 Home quality	Verify one KPI row only, non-duplicated operational stage counts, search/Copilot separation, smooth robot asset background, bottom Quick Actions and responsive layout.

20.16 Regulatory and implementation boundaries

The organisation/GST model is a product data architecture, not tax or legal advice. GST registration status, place-of-business mapping, cross-GSTIN supplies, valuation, invoice, e-way bill, tax and input-credit treatment must be confirmed by the customer's GST practitioner before real transactions are processed. The manufacturing MVP captures the operational records and references required for traceability but does not replace statutory accounting/GST filing or generate statutory tax documents unless separately scoped and approved.

Official reference points used for this clarification:

- CGST Act, section 25 — registration, including multiple registrations and treatment of each registration as a distinct person: https://taxinformation.cbic.gov.in/content-page/explore-act/1000294/1000001
- GST Portal registration guide — existing GST registrations mapped to the same PAN: https://tutorial.gst.gov.in/userguide/registration/Apply_for_Registration_Normal_Taxpayer.htm
- CBIC GST registration rules — registration certificate and principal/additional places of business: https://cbic-gst.gov.in/hindi/gst-registration-rules.html

These references support the distinction between legal entity/PAN, GSTIN registration and physical place of business. Implementation decisions must still be validated against the customer's actual GST registrations and advice from a qualified GST practitioner.

20.17 Clarification change log

Change ID	Clarification	PRD areas affected
CL-01	Defined Tenant → BusinessGroup → Company/LegalEntity → GST registrations → Sites → StoreLocations; added cross-GSTIN transfer traceability.	5, 6, 9, 10, 13, 14
CL-02	Separated Site from StoreLocation and clarified GST registration mapping.	6, 9, 13
CL-03	Added dedicated CustomerAccount and Customer Directory UX-02a.	6, 7.2, 9, Appendix A
CL-04	Clarified finished goods are Inventory; Assets are durable operational equipment.	7.5, 7.8, 8, 9
CL-05	Added customer PO PDF/image/photo evidence and immutable source/version capture.	7.2, 7.10, 8, 13
CL-06	Strengthened SupplierQuotation record/evidence/comparison requirements.	7.4, 8.2, 9
CL-07	Expanded JobMaterialRequirement into job/stage material planning and issue/return/consumption traceability.	7.6, 8.2, 8.4, 9
CL-08	Added ready-stock/partial-stock fulfilment path and direct dispatch without unnecessary production.	7.5, 7.9, 8, 9
CL-09	Clarified PO approval as configurable/conditional by ApprovalPolicy.	7.4, 10, 18
CL-10	Strengthened AuditEvent and standard created/updated actor fields; database-persisted audit history.	9, 10, 14, 15, 18
CL-11	Added MOD-01–MOD-12 to UX-01–UX-15 map and updated screen baseline to 27.	5, 6, Appendix A
CL-12	Approved Aurora Indigo Home theme across all screens; separated global search from Copilot; added robot	7.1, 11, 12, 16

Change ID	Clarification	PRD areas affected
	background and bottom-space rules.	

20. Product Review Feedback Updates (v1.3)

This section documents binding updates from the product review session of 5 November 2026 (feedback items 3–21).

20.1 Role × Module Access Control Matrix (Items 3 & 5)

Access levels: — = No Access • View = Read-only • Edit = Read+Write • Admin = Full (including config/delete)

Module	Owner
MOD-00 Home	Admin
MOD-02 Sales	Admin
MOD-03 Suppliers	Admin
MOD-04 Procurement	Admin
MOD-05 Inventory	Admin
MOD-06 Jobs	Admin
MOD-07 Quality	Admin
MOD-08 Assets	Admin
MOD-09 Dispatch	Admin
MOD-10 Documents	Admin
MOD-11 Reports	Admin
MOD-12 Admin	Admin

Item 5 — Multi-Role Users: A user may hold multiple roles simultaneously. The system grants permissions as the union of all assigned roles. The role field on user records accepts an array of role keys.

20.2 Asset Tag Numbering & Barcode System (Item 4)

Every physical asset (machines, furniture, electronics, tools) is assigned a unique tag at commissioning.

Format: EOSS-ASSET-{SITE_CODE}-{SEQUENCE} — Example: EOSS-ASSET-HYD-0042

Asset categories: MACH (machinery) • TOOL (hand/power tools) • FURN (furniture & fixtures) • ELEC (electronic devices) • VEHI (vehicles/forklifts) • INFRA (racks, shelving)

Barcode scanning actions:

- Assets screen → auto-open asset record + pre-fill Log Event modal
- Inventory screen → auto-locate material and open lot details
- Job Stage Update → record input/output material with lot traceability

Future roadmap: mobile app camera scan. QR code content = JSON {type, id, assetId}.

20.3 Global Admin / Super-Admin Console (Item 6 — v2.0 Roadmap)

Planned for v2.0 — not implemented in v1.x. Capabilities:

- Manage multiple company instances from one console

- Global configuration templates (roles, approval policies, numbering series)
- Cross-company reporting and consolidated P&L
- Multi-tenant user management and SSO configuration
- Billing and subscription management

20.4 Invoice Lifecycle & Maintenance Cycle (Item 7)

4.4 Invoice Lifecycle

Draft → Sent → Acknowledged → Partial Payment → Fully Paid. Overdue status triggers when payment is not received by due date.

Aging Buckets (AR & AP):

Bucket
Current
Early
Moderate
Late
Critical

4.5 Maintenance Cycle

Planned PM → Triggered (meter/date) → Work Order Created → In-Progress → Completed → PM Interval Reset → Next PM Scheduled.

Each completion records: technician, actual hours, parts used, next service date. Missed PMs automatically escalate asset status to pm_overdue.

20.5 Multi-Site / Multi-GST Architecture (Item 8)

One legal entity may operate multiple sites (factories, warehouses, branch offices). Each site has:

- Its own GSTIN (for intra-state vs inter-state GST compliance)
- Its own store locations and cost centre
- Shared user database and master data (customers, suppliers, materials)
- Site-scoped inventory (stock tracked per location-site)

Inter-site stock transfers tracked via the Stock Transfer module. Financial reports can be consolidated across all sites or filtered per site.

20.6 Customer Master as Standalone Entity (Item 9)

The Customer Master (/sales/customers) is a standalone entity — not embedded inside orders. It holds: legal name, trading name, GSTIN, PAN, state, contacts, payment terms, credit limit, rating.

All sales orders, invoices, and dispatch records reference the Customer Master by ID. Changes to the Customer Master do not retroactively alter closed orders.

20.7 Finished Goods Classification (Item 10)

Completed production output is received into Inventory (Finished Goods location), NOT Assets.

- Inventory: raw materials, WIP, finished goods, consumables, packaging
- Assets: machines, tools, furniture, vehicles, infrastructure

Flow: Production Completion → QC Inspection (pass) → Inventory FG location → Dispatch to customer

20.8 UX Screen to Module Cross-Reference (Item 11)

Screen ID
UX-01
UX-02
UX-02a
UX-02b
UX-04
UX-06
UX-08
UX-11
UX-12
UX-dispatch
UX-13
UX-14

20.9 Audit Trail Requirements (Item 12)

Every create/update/delete action is logged with:

Field
id
timestamp
actor
action
scope
entityType
entityRef
source
before / after

Retention: 90 days minimum (configurable by Owner). Export: CSV with date-range filter. Access: Owner and Auditor roles only.

20.10 Customer PO Attachment & Comparison (Item 13 — FR-SALES-003)

When a customer sends their PO (by email/post/WhatsApp), it must be attached to the EOSS sales order as a file (PDF/image/scan).

The system must compare the attached customer PO against the accepted quotation and highlight deviations in:

- Quantity per line item
- Rate / price
- Specifications / drawing revision
- Delivery date

• Payment terms

Deviations are presented as a diff table. The user must acknowledge each deviation before proceeding to job creation.

20.11 One PO → Multiple Jobs (Item 14 — FR-SALES-003, FR-JOB-001)

A single customer PO may be fulfilled by multiple production Jobs (batches).

Example: PO for 100 units → Job J-001 (40 units, Nov) + Job J-002 (40 units, Dec) + Job J-003 (20 units, Jan). PO remaining qty tracked automatically.

Each Job carries: linkedPoId, orderedQty, remainingPoQty. A Job may also partially fulfil its own target quantity. Dispatch uses only goodQty from QC-passed jobs.

20.12 Supplier Quotation Tracking Confirmed (Item 15 — FR-PROC-001)

Supplier quotations ARE tracked in EOSS (confirmed v1.3). Flow: PR created → sent to 2–3 suppliers → quotes received → comparison → lowest qualifying quote selected → converted to Supplier PO.

Quote comparison includes: unit price per line, total value, lead time (days), payment terms, supplier quality history score.

20.13 Partial Delivery & Ready Stock Dispatch (Item 16 — FR-DISP-001)

Partial Delivery: A dispatch can deliver a subset of ordered items. The system tracks dispatchedQty vs orderedQty and shows remaining balance. Status = partial until all quantity shipped, then complete.

Ready Stock Dispatch: If finished goods exist in inventory (FG location), the system shows a Ready to Dispatch indicator. Dispatch personnel may initiate immediately without waiting for new production. The system reserves existing FG stock against the dispatch order.

20.14 Job Material Requirements Workflow (Item 17 — FR-JOB-001, FR-INV-001)

Job Created → Material Requirements calculated (BOM × qty) → Check Inventory per material:

• Available (onHandQty ≥ required): Reserve stock → Production starts

• Shortage (onHandQty < required): Create PR → Procurement → GRN received → Reserve newly received stock → Production starts

Quantities displayed per material line:

Field
onHandQty
reservedQty
availableQty
qcHoldQty
inTransitQty
shortage

20.15 Auto-Approval Rules for Procurement (Item 18 — FR-PROC-001)

Not all procurement items require manual approval:

Condition
PR value < ₹10,000
Repeat purchase, approved vendor, standard item

New vendor
New item not in item master
Value ≥ ₹10,000, non-repeat
Value ≥ ₹1,00,000

Auto-approved PRs are flagged Auto in the approvals log.

20.16 Production Quantity Buffer (Item 19 — FR-JOB-001)

Jobs may be created for more than the PO quantity to account for expected quality rejects.

Buffer % is configurable per job (default: 5%). plannedQty = orderedQty × (1 + buffer/100).

After production: good qty dispatched to customer; rejected units → NCR/scrap; rework units re-enter production. Excess good units (beyond PO qty) go to FG inventory for future orders.

20.17 Role Count Corrected to 12 (Item 20)

Section 10.1 previously listed 10 roles. The correct count is 12 roles. The complete role list (all 12) is documented in Section 10 and reproduced below:

#
1
2
3
4
5
6
7
8
9
10
11
12

20.18 AI Copilot Guardrails (Item 21)

The following guardrails govern the EOSS AI Copilot:

1. Advisory only — Copilot cannot execute state-changing actions. All suggestions require human confirmation.
2. Role-scoped visibility — Copilot only surfaces data the current user's role can access.
3. No persistent memory — Session-only context. No queries stored after session end.
4. Hallucination disclaimer — All suggestions labelled “AI suggestion — verify before acting.”
5. Financial data access — Detailed financials only for Owner and Accounts roles.
6. Rate limits — 50 AI queries per user per hour (configurable by Owner).
7. Audit logging — Every Copilot query logged with user, timestamp, and query text.

Prototype note: responses are simulated (no live LLM API). All suggestions labelled [AI — verify before acting]. No data transmitted to external AI provider in prototype mode.